TestDirector™

Tutorial

Version 5.0

Online Guide

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Table of Contents

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Welcome to the TestDirector Tutorial

Welcome to the TestDirector Tutorial, a self-paced guide which teaches you how to use TestDirector, Mercury Interactive's process-based test management system tool.

This tutorial instructs you on how to use TestDirector to manage the software testing process. It familiarizes you with creating databases, planning and building tests, running tests, and tracking software defects.

TestDirector integrates with Mercury Interactive testing tools (WinRunner, LoadRunner, Visual API, Astra QuickTest, QuickTest 2000, and XRunner), as well as with third-party and custom testing tools.

This tutorial is designed for use with WinRunner.

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Before You Begin...

In order to use this tutorial, you need the following components installed on your computer:

✔ TestDirector 5.0

✔ TestDirector_Demo project—a database provided with TestDirector containing manual and automated sample tests.

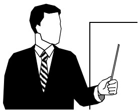
✔ WinRunner 5.0 or higher—Mercury Interactive's automated GUI testing tool.

✔ Flight Reservation application (versions 1A and 1B)—a sample application provided with WinRunner. WinRunner's typical setup, installs the sample application automatically. If you choose WinRunner's custom setup, select the samples component, in order to install the Flight Reservation application.

If you have problems running any of these components, consult your system administrator or Mercury Interactive's customer support.

What’s in this Tutorial?

The TestDirector tutorial is divided into 7 lessons:



Lesson 1: Introducing Testing

Provides you with an overview of the TestDirector workflow and familiarizes you with the TestDirector user interface.

Lesson 2 : Getting Ready to Test

Shows you how to create a master test plan based on your goals, resources, and time constraints.

Lesson 3 : Creating a New Project

Shows you how to create a new database and customize it to meet the requirements of your application.

Lesson 4: Creating a Test Plan

Shows you how to analyze the sample application and break it down into subjects. It also helps you build a test plan tree that represents the hierarchical relationship of the subjects.

Lesson 5: Designing Tests

Shows you how to define test steps. It also helps you decide whether to automate a test or perform it manually.

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Lesson 6: Running Tests

Shows you how to organize test execution by building test sets, and how to execute manual and automated tests.

Lesson 7: Tracking Defects

Shows you how to report and track defects detected in an application.

As you work through the lessons, keep in mind that you are learning the basics for managing any project. Each lesson should take approximately 20 minutes.

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TestDirector Documentation and Other Resources

Mercury Interactive offers printed and online documentation, online customer support, and other resources that will help you use TestDirector as effectively as possible.

User Guides

TestDirector User’s Guide

Provides step-by-step instructions on how to use TestDirector. It describes many useful test management tasks and options not covered in this tutorial.

TestDirector Installation Guide

Explains how to install TestDirector and the client software needed to connect TestDirector to project databases.

TestDirector Administrator’s Guide

Describes the tasks performed by a TestDirector administrator.

TestDirector Open Test Architecture Guide

Explains how to connect third-party and custom testing tools to TestDirector, and how to integrate your applications with TestDirector project databases.

Remote Defect Reporter User’s Guide

Explains how to set up and use the Remote Defect Reporter.

Visual API Guide

Explains how to use Visual API, TestDirector’s API execution tool.

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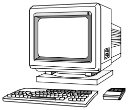
Online Resources



Read Me First

Provides current news and information about TestDirector.

TestDirector Quick Preview

Provides you with a brief overview of key TestDirector concepts and features.

TestDirector Online Help

Provides quick answers to questions that arise as you work with TestDirector.

TestDirector Books Online

Provides you with easy access to all the TestDirector books.

Note that in order to view the online books, you must first install the Acrobat Reader 3.01. To install the Acrobat Reader, select the Acrobat Reader Setup icon in the TestDirector program group.

Technical Support Online

Opens Mercury Interactive's Customer Support web site in your default browser.

Mercury Interactive on the Web

Opens Mercury Interactive's home page. This site provides you with the most up-to-date information on Mercury Interactive and its products. This includes new software releases, seminars and trade shows, customer support, educational services, and more. Visit Mercury Interactive's web site at http://www.merc-int.com.

Introducing TestDirector

Software testing is a complex process. TestDirector helps you organize and manage all phases of the software testing process, including planning, creating tests, executing tests, and tracking defects.

This lesson includes:

Why Test Your Software?

Even the most carefully planned and designed software, cannot possibly be free of defects. Your goal as a quality engineer is to find these defects. This requires creating and executing many tests.

In order for software testing to be successful, you should start the testing process as soon as possible. Each new version must be tested in order to ensure that "improvements" do not generate new defects. If you begin testing only shortly before an application is scheduled for release, you will not have time to detect and repair many serious defects. Thus by testing ahead of time, you can prevent problems for your users and avoid costly delays.

Managing the Testing Process

TestDirector simplifies test management by helping you organize and manage all phases of the software testing process, including planning, creating tests, executing tests, and tracking defects. With TestDirector, you maintain a *project*—a database of tests. From a project, you can build *test sets*—groups of tests executed to achieve a specific goal. For example, you can create a test set that checks a new version of the software, or one that checks a specific feature. As you execute tests, TestDirector lets you report defects detected in the software. Defect records are stored in a database where you can track them until they are resolved in the software.

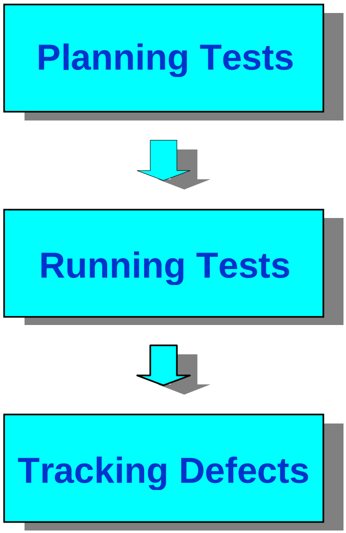
TestDirector works together with **WinRunner, Mercury Interactive's automated GUI Testing tool. WinRunner** enables you to create and execute automated test scripts. You can include WinRunner automated tests in your project, and execute them directly from TestDirector. TestDirector activates WinRunner, runs the tests, and displays the results. For details on creating automated tests in WinRunner, refer to the *WinRunner Tutorial* and the *WinRunner User's Guide*.

Besides the TestDirector and WinRunner integration, TestDirector offers integration with other Mercury Interactive testing tools (LoadRunner, Visual API, Astra QuickTest, QuickTest 2000, and XRunner), as well as with third-party and custom testing tools.

This tutorial is designed for use with WinRunner.

Understanding the TestDirector Workflow

The TestDirector workflow consists of 3 main phases:



In each phase you perform several tasks:

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Planning Tests

Divide your application into test subjects and build a project.

1 **Define your testing goals.**

Examine your application, system environment, and testing resources to determine what and how you want to test.

2 **Define test subjects.**

Define test subjects by dividing your application into modules or functions to be tested. Build a test plan tree that represents the hierarchical relationship of the subjects.

3 **Define tests.**

Determine the tests you want to create and add a description of each test to the test plan tree.

4 **Design test steps.**

Break down each test into steps describing the operations to be performed and the points you want to check. Define the expected outcome of each step.

5 **Automate tests.**

Decide whether to perform each test manually or to automate it. If you choose to perform a test manually, the test is ready for execution as soon as you define the test steps. If you choose to automate a test, use WinRunner to create automated test scripts in Mercury Interactive's Test Script Language (TSL).

6 Analyze the test plan.

Generate reports and graphs to help you analyze your test plan. Determine whether the tests in the project will enable you to successfully meet your goals.

Running Tests

Create test sets and perform test runs.

1 Create test sets.

Create test sets by selecting tests from the project. A test set is a group of tests you execute to meet a specific testing goal.

2 Run test sets.

Schedule test execution and assign tasks to testers. Run the manual and/or automated tests in the test sets.

3 Analyze the testing progress.

Generate reports and graphs to help you determine the progress of test execution.

Tracking Defects

Report defects detected in your application and track how repairs are progressing.

1 **Report defects.**

Report defects detected in the software. Each new defect is added to the defect database.

2 **Track defects.**

Review all new defects reported to the database and decide which ones should be repaired. Test a new version of the application after the defects are corrected.

3 **Analyze defect tracking.**

Generate reports and graphs to help you analyze the progress of defect repairs, and to help you determine when to release the application.

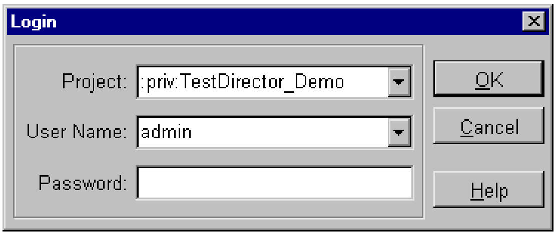
Exploring the TestDirector Window

Before you begin the testing process, you should familiarize yourself with the main TestDirector window.

To open TestDirector:

1 **Start TestDirector.**

Choose TestDirector from the TestDirector 5.0 program group.



2 **Select a project.**

In the Login window, select :priv:TestDirector_Demo from the Project list.

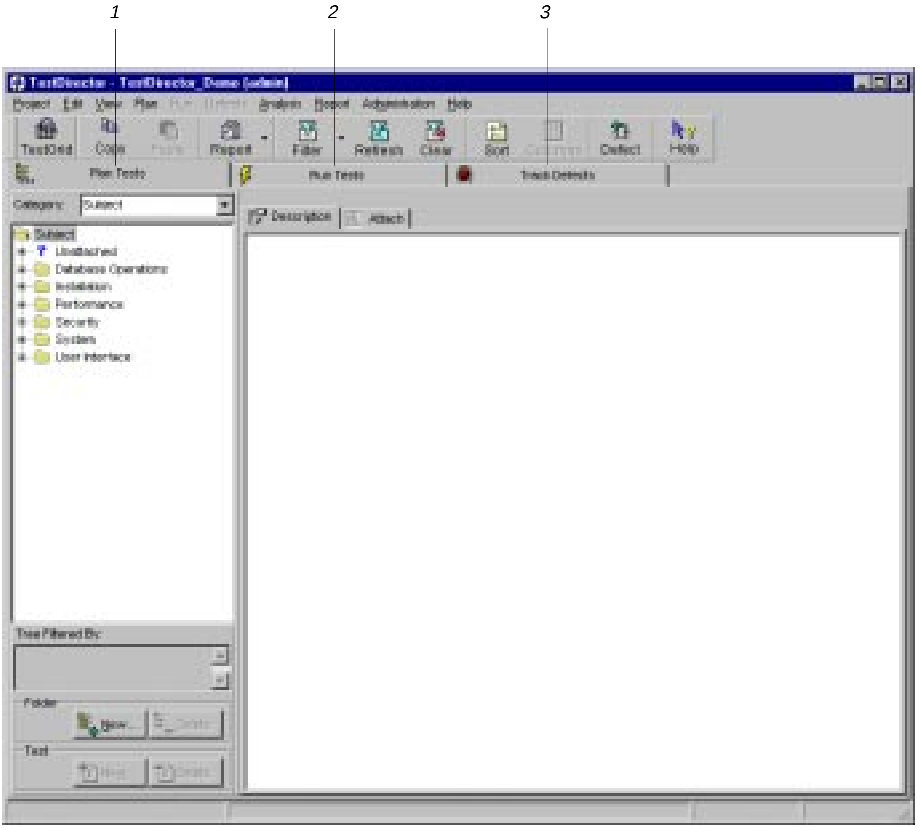
The :priv: prefix indicates that the TestDirector_Demo project is stored in your private directory. It can only be accessed by you.

A project without the :priv: prefix indicates that it is stored in a common directory. Unlike your private directory, this project can be accessed by all users.

3 **Select a user name.**

For the purposes of this tutorial, select the user name admin. You do not need to type a password. Click OK to open the TestDirector window.

In TestDirector, you can work in three separate views, according to the tab you click in the main TestDirector window.

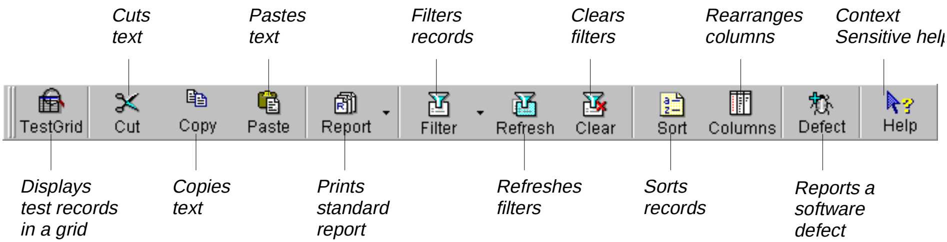


1 *Plan Tests tab: Create a test plan tree and plan tests for each subject in the tree.*

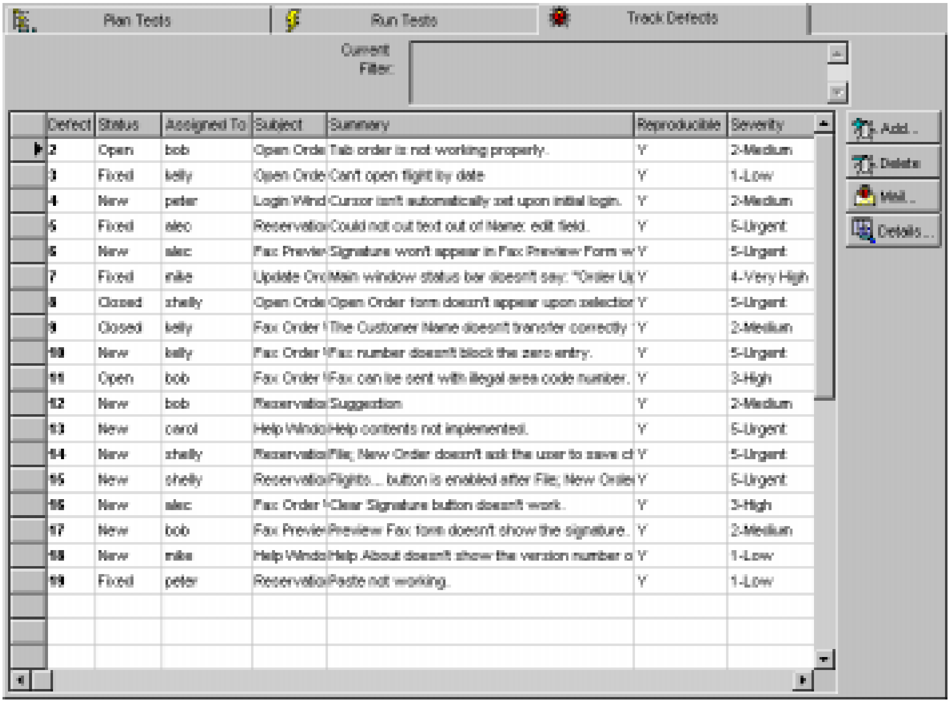
2 *Run Tests tab: Create test sets and execute manual and automated tests.*

3 *Track Defects tab: Report and track defects detected in your application.*

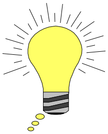
The toolbar is common to all TestDirector views. It provides easy access to frequently performed test management tasks, such as editing project records, printing status reports, and reporting software defects.



TestDirector organizes and displays data in grids. Using the grids, you can view and modify information in your project. In the following example, the Track Defect grid displays the defect reports in a TestDirector project. This grid appears when you select the Track Defects tab.



Now that you are familiar with the TestDirector workflow and TestDirector's main window, you can proceed to Lesson 2, "**Getting Ready to Test**." In Lesson 2, you will define your testing goals.



Tips

- You can determine which columns appear in a grid and the order in which they appear by choosing View > Columns. The Column Select/Order dialog box opens.
- You can resize grid columns using the mouse. Click the right edge of a column and drag to adjust the width. You can only resize columns that are not fixed.

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Getting Ready to Test

Before you begin testing your application, you should define your testing goals and outline a strategy for achieving them. Start by closely examining the application you plan to test.

This lesson includes:

- **Touring the Flight Reservation Application**
- **Defining Your Testing Goals**

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Touring the Flight Reservation Application

The Flight Reservation application is a client/server application designed for travel agents. An agent can book flights, fax reservations, and produce reports and graphs summarizing reservation information. In order to use the system, the travel agent is required to log on using a name and password.

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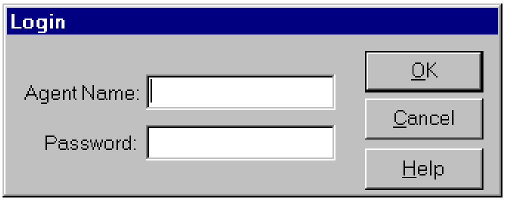
In this exercise you will explore the Flight Reservation application—a sample application provided with WinRunner.

To examine the Flight Reservation application:

1 **Open version 1A of the Flight Reservation application.**

Choose Flight 1A from the Sample Applications submenu in the WinRunner program group.

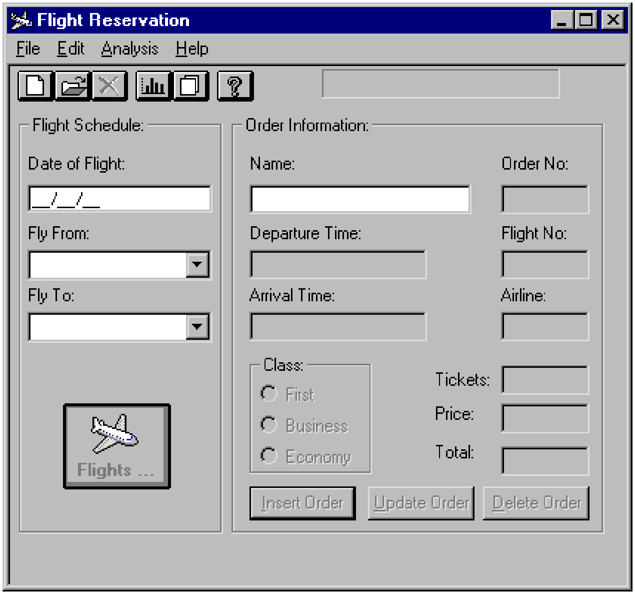


2 **Log on using your name.**

Type your name in the Agent Name box (using at least 4 characters).

3 Type a password.

Type mercury in the Password box and click OK. The Flight Reservation window opens.



4 Create a new order.

Choose File > New Order in the Flight Reservation window.

5 Type the date of flight.

Type the date 11/11/98 in the Date of Flight box.

As you fill in the fields, notice what information you have to enter before each of the command buttons (Flights, Insert Order, Update Order, Delete Order) is enabled.

6 Select the flight route.

Select Los Angeles from the Fly From list.

Select San Francisco from the Fly To list.

7 Select a flight.

Click Flights and select flight 4219 from the Flight Table. Click OK.

8 Type customer name.

Type the name David Johnson in the Name box.

9 Select a flight class.

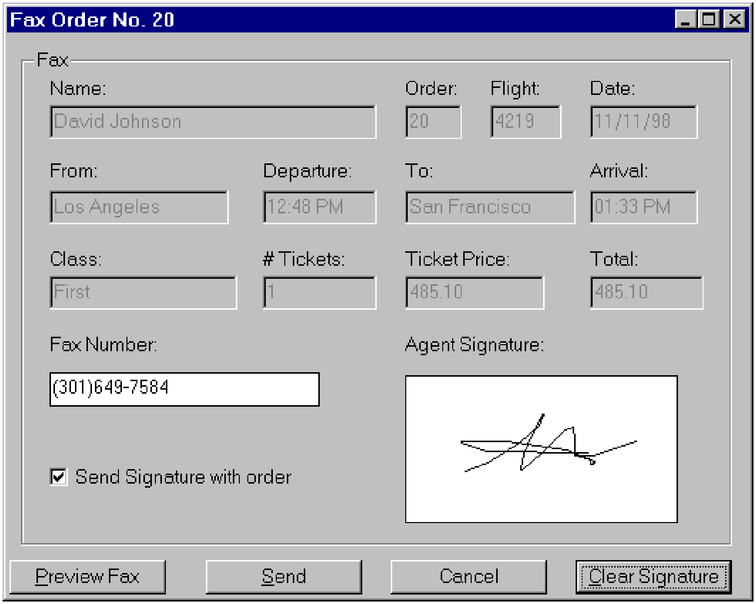
Select First.

10 Insert the order into the database.

Click Insert Order.

11 Specify fax order information.

Choose File > Fax Order. In the Fax Number box, type (301)649-7584. In the Agent Signature box, sign your name. Select “Send Signature with order.”

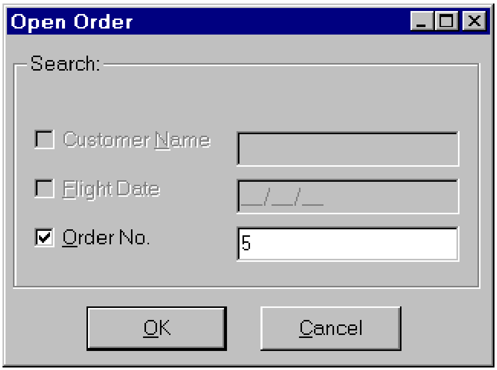


12 Transmit the fax order.

Click Preview Fax and then click Send. The Fax Order dialog box closes. The “Fax Sent Successfully...” message appears in the message bar of the Flight Reservation window.

13 Open order number 5.

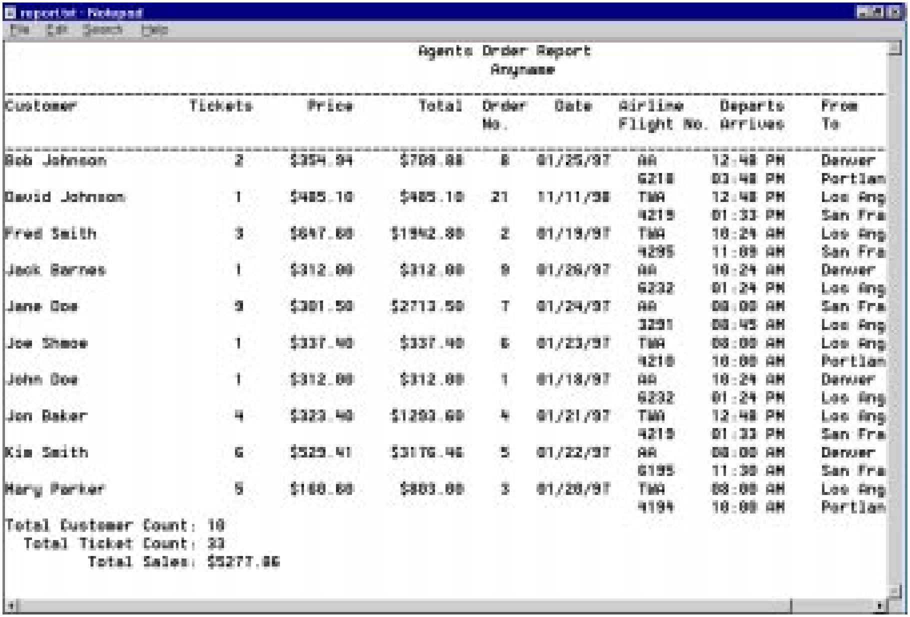
Choose File > Open Order. In the Order dialog box, click the Order No. checkbox. In the adjacent box type 5, and click OK.



Order number 5 appears in the Flight Reservation window.

14 Generate a report to analyze the contents of the database.

Choose Analysis > Reports to view a report of all the records in the database. Choose File > Exit to close the report.


report.txt - Notepad

File Edit Search Help

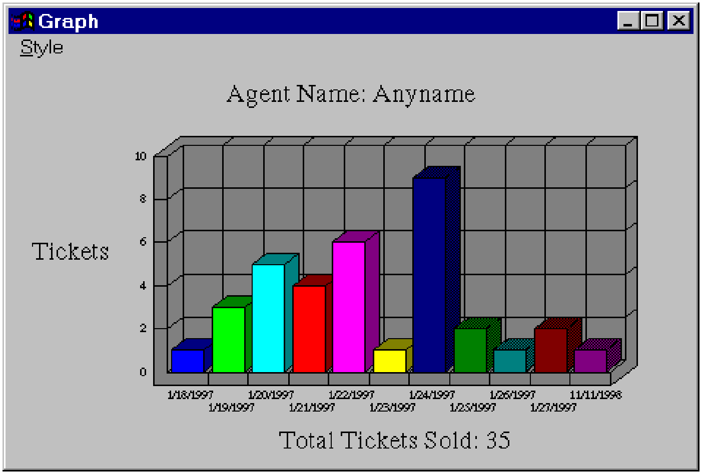
Agents Order Report
Anyname

Customer	Tickets	Price	Total	Order No.	Date	Airline Flight No.	Departs Arrives	From To
Bob Johnson	2	$354.94	$709.88	8	01/25/97	AA	12:48 PM	Denver
						6218	03:48 PM	Portlan
David Johnson	1	$485.10	$485.10	21	11/11/98	TWA	12:48 PM	Los Ang
						4215	01:33 PM	San Fra
Fred Smith	3	$647.60	$1942.80	2	01/19/97	TWA	10:24 AM	Los Ang
						4295	11:09 AM	San Fra
Jack Barnes	1	$312.00	$312.00	9	01/26/97	AA	10:24 AM	Denver
						6232	01:24 PM	Los Ang
Jane Doe	9	$301.50	$2713.50	7	01/24/97	AA	08:00 AM	San Fra
						3291	08:45 AM	Los Ang
Joe Shmoe	1	$337.40	$337.40	6	01/23/97	TWA	08:00 AM	Los Ang
						4210	10:00 AM	Portlan
John Doe	1	$312.00	$312.00	1	01/18/97	AA	10:24 AM	Denver
						6232	01:24 PM	Los Ang
Jon Baker	4	$323.40	$1293.60	4	01/21/97	TWA	12:48 PM	Los Ang
						4215	01:33 PM	San Fra
Kim Smith	6	$529.41	$3176.46	5	01/22/97	AA	08:00 AM	Denver
						6195	11:30 AM	San Fra
Mary Parker	5	$160.60	$803.00	3	01/20/97	TWA	08:00 AM	Los Ang
						4194	10:00 AM	Portlan

Total Customer Count: 10
Total Ticket Count: 33
Total Sales: $5277.86

15 Generate a graph to analyze the contents of the database.

Choose Analysis > Graphs see how many tickets have been sold and for which dates. Press Alt+F4 to close the graph.



Defining Your Testing Goals

Once you are familiar with the application you plan to test, you should define your testing goals by examining your testing environment, resources, and constraints. Consider the following:

Examine your system environment

Find out:	For example:
What type of database server are you using?	Oracle, Sybase, Microsoft SQL, Microsoft Access
Which networks?	Novell, LAN
Which platforms?	Windows 95/NT, UNIX, OS/2
Which languages?	English, Japanese

Examine your testing resources

Find out:	For example:
What computers are available?	5 PCs, 4 UNIX workstations
How much time do you have to test?	3 months
What skills do your testers have?	3 test engineers, 5 college students

Determine management priorities

Find out:	For example:
What is the goal of the upcoming software release?	Create a stable beta version
What is the desired release date?	October 15

Identify the risks if the software malfunctions

Find out:	For example:
What types of failures could cause the most damage?	Failure to process ticket orders. Failure to prevent unauthorized access.

Prioritize the testing tasks

Find out:	For example:
Which application modules will be tested first?	Login procedures, user interface

Once you have answered these questions, create a document detailing your testing goals. This document should be reviewed by the top managers, project managers, product managers, and quality assurance managers, and subsequently revised to incorporate their input.

Now that you are familiar with the application that you plan to test, and the process of defining your testing goals, you can proceed to Lesson 3, "**Creating a New Project**." In Lesson 3, you will create and customize the flight_db project.

Creating a New Project

Once you have determined your testing goals, you create a TestDirector project.

This lesson includes:

- **Creating a New Project**
- **Customizing a New Project**
- **Tips**

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Creating a New Project

You create a project for saving and retrieving information relevant to your plan. A TestDirector *project* is a database that contains test data, test steps, test runs, and defect reports. You create a project using the Project Administration utility.

In this exercise you will create the flight_db project. Note that certain procedures in this exercise discuss client/server database options. These options are available only in the TestDirector client/server edition, not in the standard edition.

To create a new project:

1 **Start the Project Administration utility and log on.**

Choose Project Administration from the TestDirector 5.0 program group.

In the Login window, the Administrator Login field is set to SYSADMIN. You do not need to type a password. Click OK.

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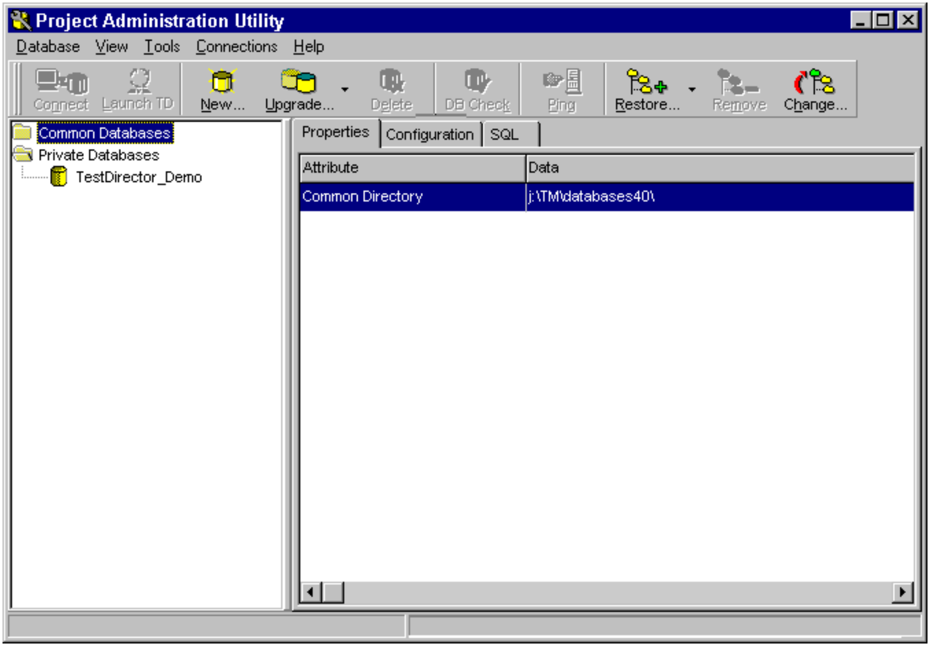
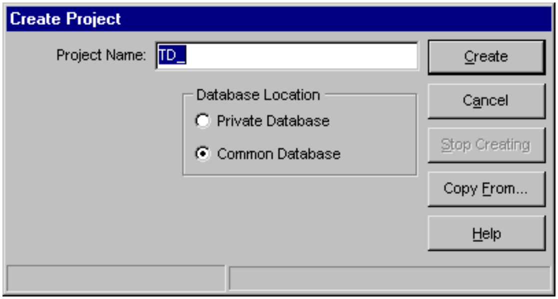


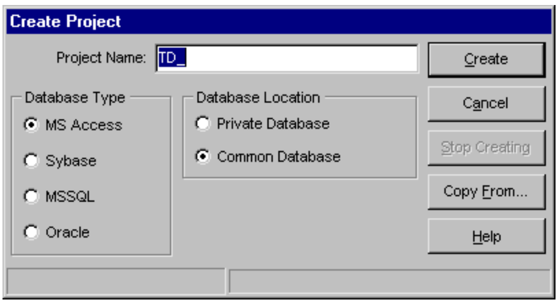
2 Open the Create Project dialog box.

Choose Project > New Project or click New.

The following Create Project dialog box opens in the standard edition of TestDirector.



The following Create dialog box appears in the client/server edition of TestDirector.



3 Type the project name.

In the Project Name box, delete the default prefix TD_ and type flight_db.

4 Choose the database type (TestDirector client/server edition only).

TestDirector enables you to work with several database types:

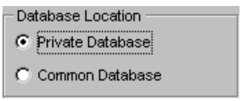
Ms Access (the default): Creates a TestDirector project that can reside on your computer, or in a common directory.

Sybase, *MSSQL*, or *Oracle:* Creates a TestDirector project that resides on a database server. When the project is created, TestDirector automatically creates a subdirectory in the common directory.

For the purposes of this tutorial, select MS Access.

5 Choose the database location.

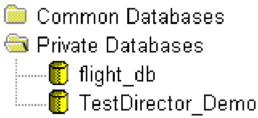
You can choose to share the new project with many users (common), or to create it for your personal use only (private). For the purposes of this tutorial, select Private Database.



6 Create the flight_db project.

Click Create. A message appears stating that the new project is created successfully. Click Ok.

The flight_db project is added to the Private Databases list in the Project Administration window.



7 Close the Project Administration utility.

Choose Project > Exit.

Customizing a New Project

After you create a new project, customize it to meet the needs of your testing environment:

✔ List the users who will have access to the project and assign each user a password and testing privileges.

✔ Add fields and selection lists to the project.

Defining Project Users

TestDirector enables you to manage user access to a project. By creating a list of authorized users and assigning each user a password and a user group, you control the kinds of additions and modifications each user can make to the project. The user group determines the privileges that the user has within TestDirector. You can assign users to custom or standard user groups. Following are the standard TestDirector user groups and their default privileges:

- *TDAdmin*: Has full privileges in a TestDirector project. This is the only user type that can change user and group privileges, or customize a project.
- *QATester*: Create, modify, and delete tests and subjects in Plan Tests mode. Users with QA Tester status can also create test sets, run tests, delete test runs, and report defects in Run Tests mode. They can also change defect status from *Fixed* to any other status.
- *Project Manager*: Report new defects, delete defects, and modify defect status.

- *Developer*: Report new defects and change defect status to *Fixed* or *Rejected*.
- *Viewer*: Has read-only privileges in a project database.

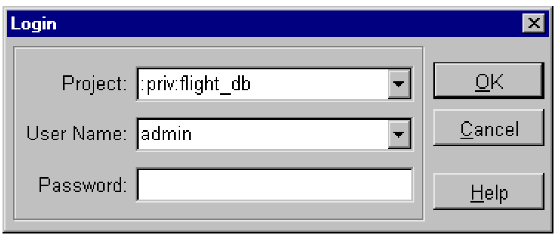
All TestDirector projects include two default user types: *admin* and *guest*. The admin user has the privileges of a TestDirector Administrator; the guest user has Viewer privileges. You cannot assign these users to a different user group, nor can you delete these users from a project database.

In this exercise you will define two users who will be authorized to work with the flight_db project. For each user, you will assign a password and a user group.

To add users:

1 Open the flight_db project.

If TestDirector is not already open, choose TestDirector from the TestDirector 5.0 program group. The Login window opens. Select flight_db from the Project list. Select admin from the User Name list. Click OK.



If TestDirector is already open, choose Project > Open. The Login window opens. Select flight_db from the Project list. Select admin from the User Name list. Click OK.

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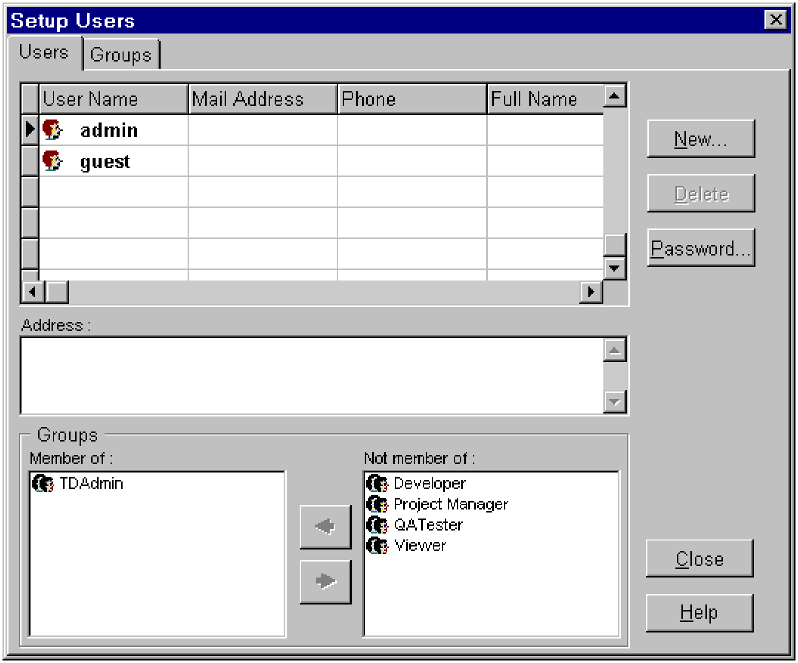
2 Open the Administrator window.

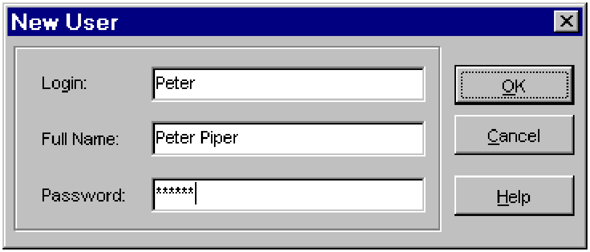
Choose Administration > Setup Users.

By default, the Users tab lists two users: admin and guest. Admin is assigned TDAdmin privileges and guest is assigned Viewer privileges.



3 **Add Peter Piper to the user list.**

Click New. In the Login box type Peter and in the Full Name box, type Peter Piper. In the Password box, type pepper.



Click OK. Peter Piper is added to the User list.

4 **Assign Peter to the QATester user group.**

In the Users table, click the name Peter. Notice that in the “Members of” list, Peter is listed as a Viewer. This is the default user type for new users.

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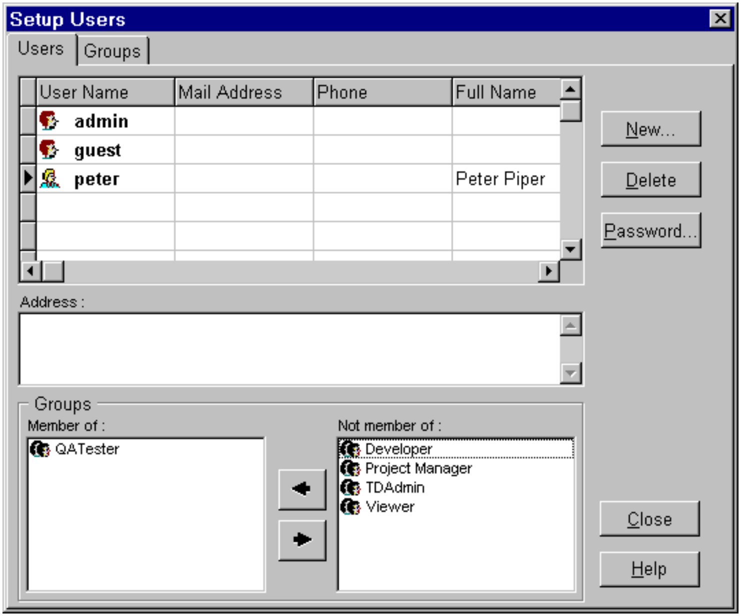
Select Viewer and click the arrow to move this user type to the “Not Member of” list. In the “Not Member of” list, select QATester and click the arrow. Peter is now assigned QATester privileges.



5 Add Kelly Jones to the user list.

Click New. The New User dialog box appears. In the Login box type Kelly and in the Full Name box type Kelly Jones. In the Password box type wizard7.

Click OK. Kelly Jones is added to the User list.

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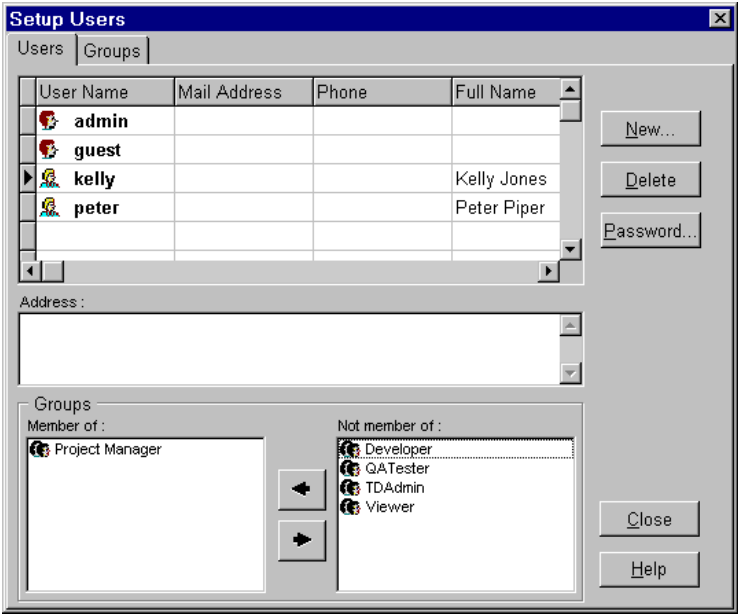
6 Assign Kelly to the Project Manager user group.

In the users table, click the name Kelly. In the “Members of” list, select Viewer and click the arrow. In the “Not Member of” list, select Project Manager and click the arrow. Kelly is now assigned Project Manager privileges.



7 Close the Administrator window.

Click Close.

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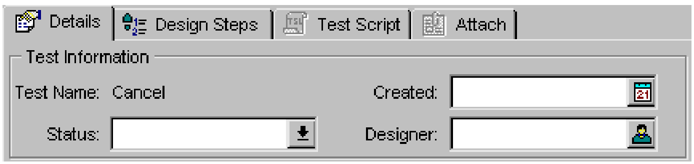
Adding Fields and Selection Lists

The new project contains standard fields for entering test data. You can customize the project by adding user-defined fields to TestDirector dialog boxes and grids. User-defined fields enable you to add information relevant to the application. For each user-defined field you can also assign a selection list containing values the user can select for the field.

For example, the Details tab in the Plan Tests view, includes 3 fields Status, Created, and Designer.



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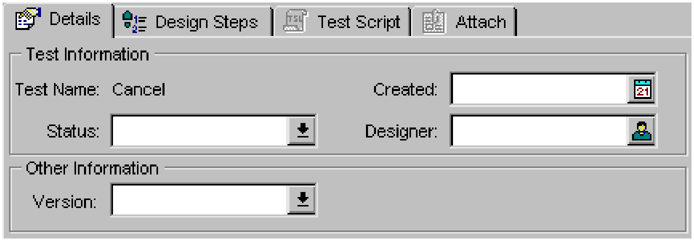
Suppose you want to add a custom field for specifying the version of your application. Your view would include 4 fields Status, Created, Designer, and Version.



Note: In order for the Details tab to appear in the Plan Test view (as in the example above), you must first create a test in your new project database. In Lesson 5, “**Designing Tests**”, you will create a test and define the test details.

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In this exercise you will add the Version field for specifying the version of the Flight Reservation application. You will also create a list of values (Flight 1A, Flight 1B, Flight 1C, Flight 1D) which can be selected for this new field.

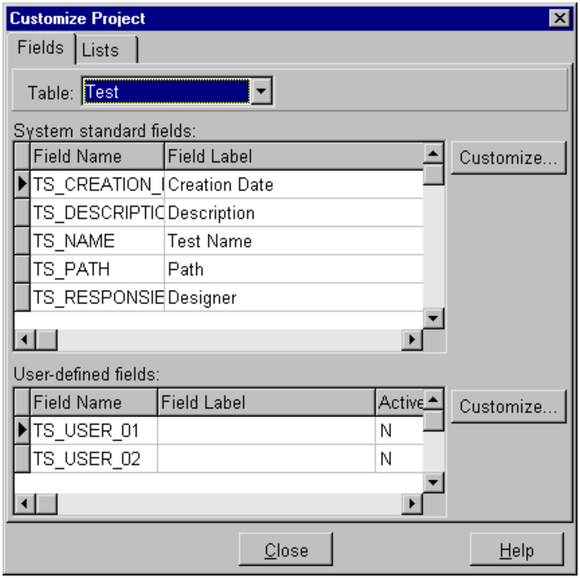
To add a new field and a selection list:

1 **Open the Customize Project window.**

Choose Administration > Customize Project.

2 **Select the table that you want to customize.**

In the Table list, select Test. Test contains all the fields used to store information about tests in your project.



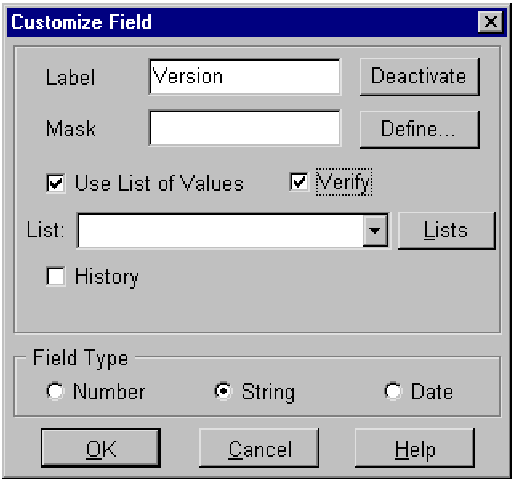
3 **Add the user-defined field Level to the Test grid.**

The user-defined field section at the bottom of the dialog box lists fields with generic names such as TS_USER_01. You create a user-defined field by assigning a label to a generic field and specifying its characteristics.

Click the field name TS_USER_01 and then click the Customize button next to the user-defined field section.

4 Assign a label name for the TS_USER_01 field.

Type Version in the Label box.



5 Create a selection list called Version.

Select the “Use List of Values” check box and the Verify check box.

Click the Lists button. The Customize Lists dialog box opens.

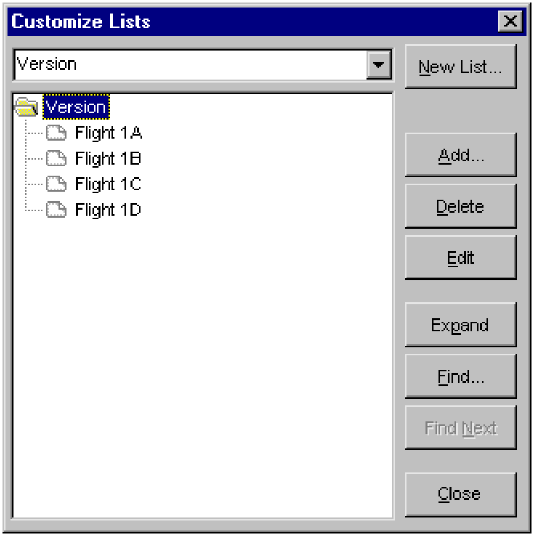
Click New List. In the New List window, type Version and then click OK.

The Version list now appears in the Customize Lists dialog box.

6 Add values to the Version selection list.

Click on the Version folder and then click Add. In the Add Folder dialog box, type Flight 1A, and click OK. Flight 1A is added to the Version selection list.

Add the values Flight 1B, Flight 1C, and Flight 1D to the Version list. Make sure that the Version folder is selected before you click the Add button.



7 Close the Customize Lists dialog box.

Click Close.

8 Attach the Version selection list to the Version field.

In the Customize Field dialog box, select Version from the List field.

9 Close the Customize Field dialog box.

Click OK. Note that Version is now listed as the field label for TS_USER_01.

10 Close the Customize Project dialog box.

Click Close. A message tells you that the changes will take affect only after you exit and reopen the project. Click OK.

11 Exit TestDirector.

Choose Project > Exit.

12 Restart TestDirector.

Choose TestDirector from the TestDirector 5.0 program group. Make sure that the flight_db project and the admin user name appear in the Login window. Click OK.

TestDirector loads and opens the customized flight_db project.

Now that you are familiar with creating and customizing a new TestDirector project, you can proceed to Lesson 4, ”**Creating a Test Plan**.” In Lesson 4, you will build a test plan tree based on a test plan for the Flight Reservation application.

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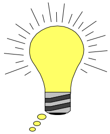
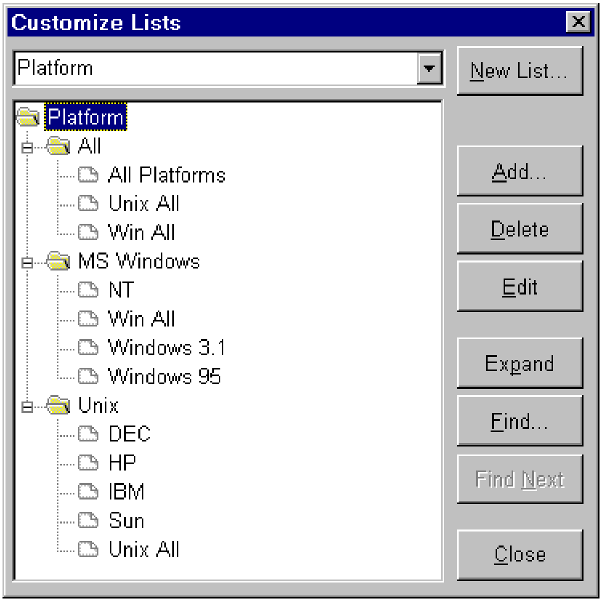


Tips

- In addition to creating new projects, the Project Administration utility lets you upgrade projects created with earlier versions of TestDirector.
- You can change your password at any time while working in TestDirector. Choose Administration > Change Password.
- Using the Customize Lists dialog box, you can organize values in a selection list by creating hierarchical levels.



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- TestDirector provides several standard selection lists. For example, it includes a “Priority” selection list containing the values: 1-Low, 2-Medium, 3-High, 4-Very High, and 5-Urgent. You can assign a standard list to any user-defined field.

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Creating a Test Plan

After you have defined your goals and customized your project, you are ready to create a test plan. A clear and concise test plan will help you evaluate the quality of your application throughout the process.

This lesson includes:

- **How Do You Create a Test Plan?**
- **Building a Test Plan Tree**
- **Tips**

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How Do You Create a Test Plan?

Developing a comprehensive test plan is an essential requirement for successful software testing. The test plan should describe the functional units or *subjects* of the application you intend to test, and the relationship between the subjects.

To create a test plan, look closely at every aspect of your application and determine its main testing subjects. Then assign each subject a name.

For example, you could define the following main subjects for the Flight Reservation application:

Main Subjects	
Database Operations	Does the software successfully perform the required operations?
Installation	Does the installation program successfully install the program?
Performance	Does the application respond to input at the expected speed?
Security	Does the log on procedure prevent unauthorized users from entering the database?

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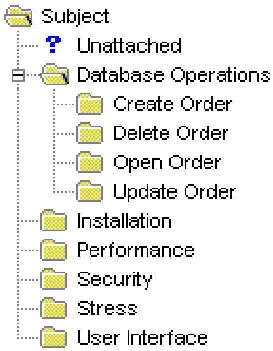
Main Subjects	
Stress	Can the application handle the stress of repeated operations?
User Interface	Does the user interface respond to input as expected?

Next, examine each main subject and break it down into smaller testing subjects. For example, you could define the following subjects under the Database Operations subject.

Database Operations	
Create Order	Does the application successfully create a new order?
Delete Order	Can the application delete an existing order from the database?
Open Order	Can the application open an existing order?
Update Order	Can the application update an existing order?

TestDirector helps you organize your test plan using a *test plan tree*. The test plan tree enables you to view the hierarchical relationship of your testing subjects. You create a test plan tree by defining a branch for each subject you plan to test.



In Lesson 5 “Designing Tests”, you will complete your test plan by designing tests and assigning them to subjects in the tree.

Building a Test Plan Tree

In this exercise you will build a test plan tree for the Flight Reservation application, based on the subjects we defined in the previous section.

To build a test plan tree:

1 **Open the flight_db project.**

If TestDirector is not already open, choose TestDirector from the TestDirector 5.0 program group. The Login window opens. Select flight_db from the Project list. Select admin from the User Name list. Click OK.

2 **Display the Plan Tests view.**

Click the Plan Tests tab.

The test plan tree is displayed on the left side of the view. The Subject folder appears at the top level of the tree. An Unattached folder appears under Subject. For more information on unattached folders, see **Tips** on page 72.

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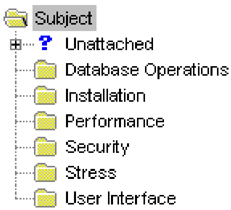
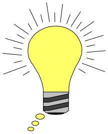
3 Add subjects to the test plan tree.

Select Subject in the tree. In the Folder area below the tree, click New. In the Add Folder dialog box, type Database Operations. Click OK. The test plan tree now includes the Database Operations subject.

Add the subjects Installation, Performance, Security, Stress, and User Interface to your test plan tree.



Now that you are familiar with defining the main testing subjects of the Flight Reservation application, and building a test plan tree, you can proceed to Lesson 5, “Designing Tests.” In Lesson 5, you will create a test based on the test plan for the Flights Reservation application.



Tips

- You can easily sort your test plan tree by name or create a custom sort by choosing View > Sort.
- You can modify the contents of a test plan tree. Activities include changing subject names, and deleting subjects and tests. Click a subject folder or a test in the tree, and choose the appropriate command from the Plan menu.
- You can create a test plan document that converts the information displayed in your test plan tree to text and exports it to a file. You can then view and edit the file. To create a test plan document, choose Report > Planning Document.

Designing Tests

After defining the subjects in the test plan tree, you are ready to design tests. You create test steps for each test. These steps describe the operations to perform, points to check, and expected results. After you define test steps, decide whether to perform the test manually or to create an automated test script.

This lesson includes:

- **Completing the Test Plan**
- **Adding a Test to the Test Plan Tree**
- **Defining Test Details**
- **Defining Test Steps**
- **When Should You Automate a Test?**
- **Analyzing the Test Plan**
- **Tips**

Completing the Test Plan

In order to complete your test plan, you create tests for each subject defined in the test plan tree. A test should check that a specific function in your application works correctly. The test you define should be based on the testing goals you determine at the beginning of the testing process. For example, in the Flight Reservation application, you could define a test called Password that checks the log on procedure. You could place this test under the Security subject.

To create a test for a subject:

✔ Define the test name in the test plan tree under the appropriate subject.

✔ Define basic test details such as the creation date and the name of the person designing the test.

✔ Define the *test steps*. These are the actions the user will take to perform the test and the expected outcome of each action.

✔ Decide whether the test should be performed manually, or automated in WinRunner.

In the following exercises your goal is to create a test called Password that checks that users cannot enter the Flight Reservation application using an illegal password. You will use the flight_db project.

Adding a Test to the Test Plan Tree

In the first stage of designing a test, you define the test name in the test plan tree.

To add a test to the test plan tree:

1 **Open the flight_db project.**

If TestDirector is not already open, choose TestDirector from the TestDirector 5.0 program group. The Login window opens. Select flight_db from the Project list. Select admin from the User Name list. Click OK.

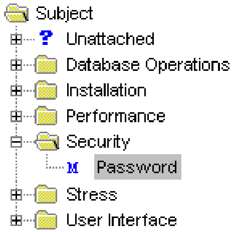
2 **Display the Plan Tests view.**

Click the Plan Tests tab.

3 **In the test plan tree, add the Password test to the Security subject.**

Select Security in the test plan tree. In the Test area below the tree, click New.

In the Create New dialog box, select the test type MANUAL, and type Password in the Test Name field. Click OK. The Password test appears in the test plan tree under Security.



Note that the icon appears next to the test name. This indicates that “Password” is a manual test. When the green footprint appears next to a test, it indicates that steps have been defined for the test. When the icon appears next to a test, it indicates that it is an automated WinRunner test.

Defining Test Details

After you add the Password test to the test plan tree, you are ready to define the basic details of the test.

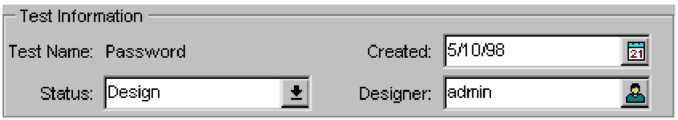
To define the test details:

1 **Display the Details view.**

Select the Password test in the test plan tree. Click the Details tab in the Plan Tests view.

2 **In the Test Info area, check that the details in the Status, Created and Designer fields are correct.**

TestDirector automatically fills in these three standard fields.

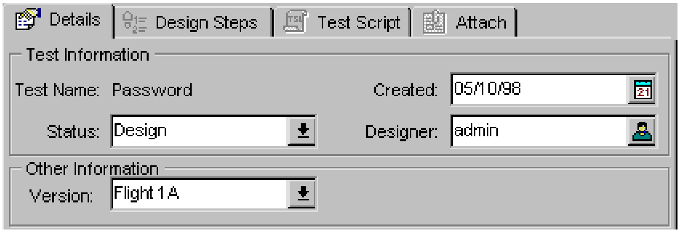


The Created field displays the current date. The Designer field displays the name of the person who created the test. The Status field displays “Design” because you have not yet designed test steps for the test.

3 In the Other Information area, assign a value to the Version field.

In the Version list, select Flight 1A. Click the check mark to close the list.

Note that the Version field is the user-defined field that you added in Lesson 3, "**Creating a New Project**", in the section **Adding Fields and Selection Lists** on page 37.



4 Type a brief test description.

In the Description box, type This test checks that users cannot enter the Flight Reservation application using an illegal password.

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Defining Test Steps

You are now ready to define test steps for the Password test. In order to check that unauthorized users cannot log on to the Flight Reservation application, the test should do the following:

- ✔ Open the Login window.
- ✔ Type a user name in the Agent Name box.
- ✔ Type an illegal password into the password box.
- ✔ Close the error message.
- ✔ Exit the Login window.

For each action the user performs in the test, you define a test step. The test step should include a brief description of the step and the expected outcome.

To define the test steps:

1 **Display the Design Steps tab in the Plan Tests view.**

Make sure the Password test is selected. Click the Design Steps tab in the Plan Tests view.

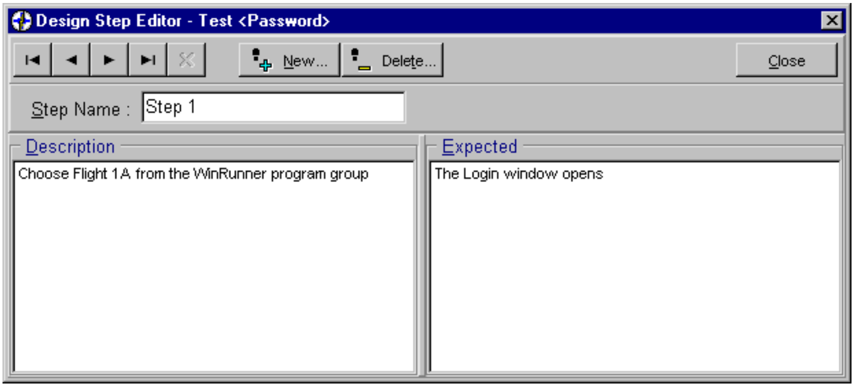
2 **Open the Design Steps Editor.**

Click New. The Design Steps Editor opens and displays “Step 1” in the Step Name box.

3 Define the first step of the test.

In the Description box, describe the first action the tester will perform. For example, type Choose Flight 1A from the WinRunner program group.

In the Expected box, describe the expected outcome of the step. For example, type The Login window opens.



4 Define the second step of the test.

Click New. Step 2 appears in the Step Name box. Define the description and expected outcome of the 2nd step.

For example, in the Description box type Type a user name in the Agent Name box. In the Expected box, type The user name appears in the Agent Name box.

5 Define the third step of the test.

Click New. Step 3 appears in the Step Name box. Define the description and expected outcome of the third step.

For example, in the Description box type Type an illegal password in the Password box and click OK. In the Expected box, type An error message appears.

6 Define the fourth step of the test.

Click New. Step 4 appears in the Step Name box. Define the description and expected outcome of the fourth step.

For example, in the Description box type Click OK to close the error message. In the Expected box, type The error message closes.

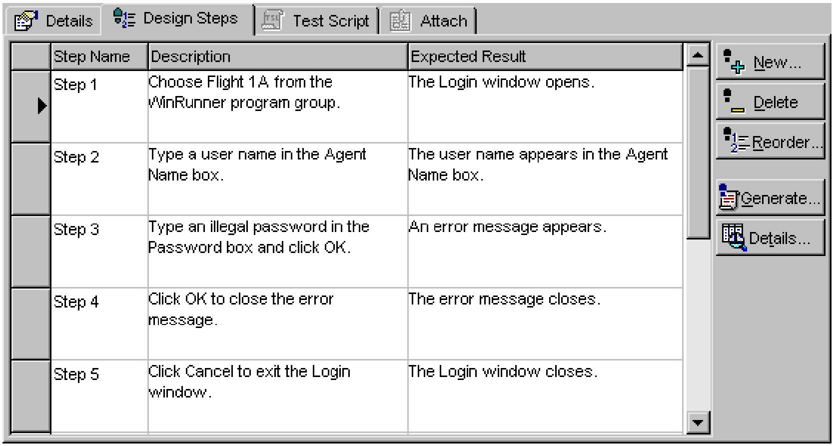
7 Define the fifth step of the test.

Click New. Step 5 appears in the Step Name box. Define the description and expected outcome of the fifth step.

For example, in the Description box type Click Cancel to exit the Login window. In the Expected box, type The Login window closes.

8 Close the Design Steps Editor.

Click Close. TestDirector displays the test steps in the Design Steps grid.



Note that in the test plan tree, a footprint now appears next to the Password test. This indicates that test steps are defined.

When Should You Automate a Test?

Once you have designed test steps, you decide whether to automate. If you choose to perform a test manually, the test is ready for execution as soon as you define the test steps.

If you choose to automate a test, allocate some additional time for creating a WinRunner test script. In many cases, the additional time you take to automate pays off later on in the testing process.

Use the following guidelines to help you decide whether to automate a test.

Do automate if:	**Do *not* automate if:**
The test is intended for repeated execution in order to “break” the application (stress test).	The test will be executed only once.
The test will be run on each new build to check the application’s basic functionality (sanity test).	The test requires immediate execution.
The test uses a different set of data each time the test is run (data-driven test).	The test does not have predictable results.
	The test is intended to check how easy the application is to use (usability test).

For example, in order for the Password test to be effective, the test should check how the log on procedure responds to many different types of illegal passwords. You could automate the Password test so that it runs many times, entering a different illegal password each time. This would be quicker and easier than manually performing many runs of the test

To automate a test, you have two options:

- In TestDirector, you can generate an automated test template for a test based on its test steps (click the Generate button in the Design Steps view). You then open the template in WinRunner and add TSL (Test Script Language) statements that perform the actions outlined in the template.
- In WinRunner, you can use recording and/or programming to create an automated test script. Afterwards, you associate the script with a test in the project database.

  For more information on automating a test, refer to the *TestDirector User's Guide* and the *WinRunner User's Guide*.

Analyzing the Test Plan

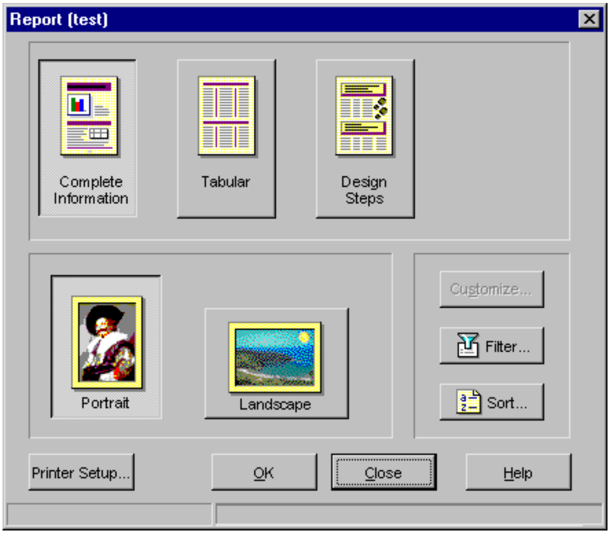
You can evaluate the progress of your test plan at any time by generating a report. A report displays information about the tests in your project database. You can generate a report that contains details about the tests and their steps, or only basic test information. You can also generate a short report that presents test information in a tabular format.

In this exercise you will generate a standard report containing test details and steps of every test in the project database.

To create a report:

1 **Open the Report dialog box.**

Choose Report > Standard, or click Report.



2 Select the Design Steps report format.

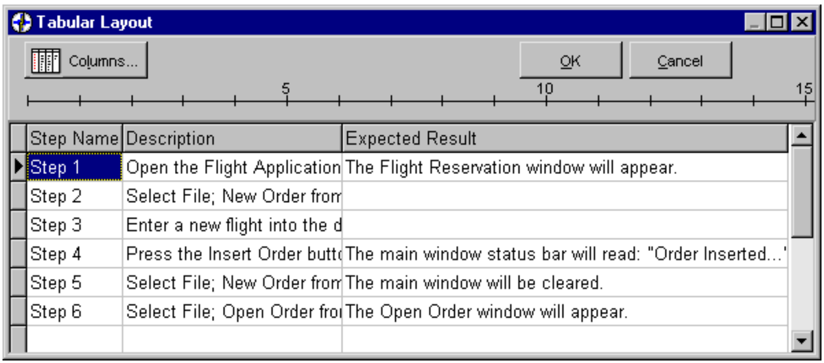
Click Design Steps. This report format includes basic details and the design steps for each test in the project.

3 Select the page layout.

Click Portrait (default).

4 Customize the report.

Click Customize. The Tabular Layout dialog box appears, displaying the information relevant to the report type that you selected.



To resize a column, place the pointer on the right edge of the column heading and move the handle to adjust the width.

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If you do not change the customization settings, the default setting for the report type will appear.

Click OK to close the dialog box.

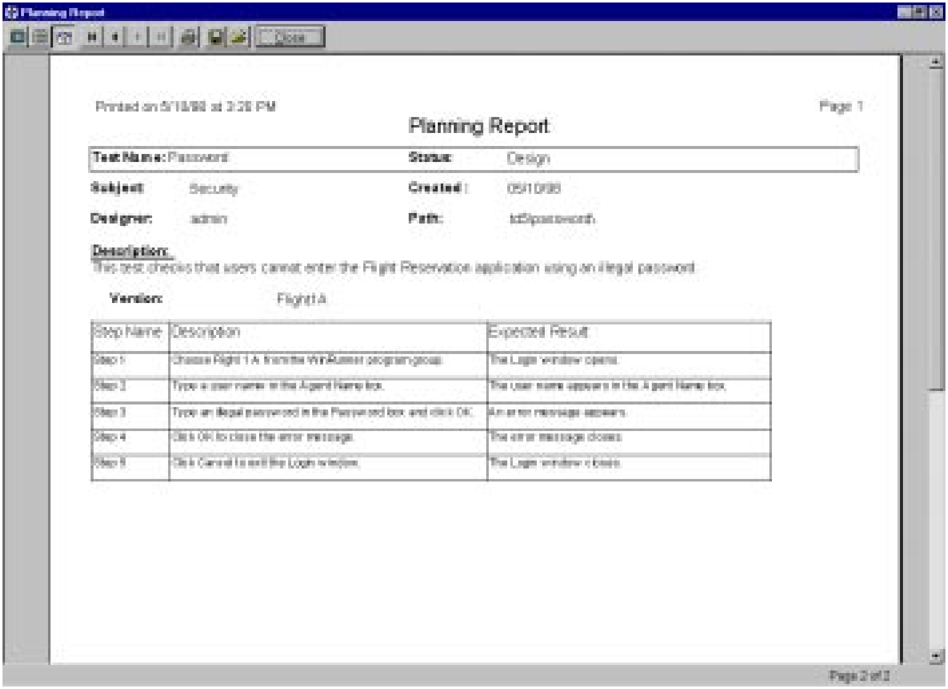
5 **Generate report.**

In the Report dialog box, Click OK. The report opens.

6 **View the report.**

Click Next Page to advance to the Password test.

The report contains the test details and test steps of the Password test that you created in the Security subject folder.



7 Close the report.

Click Close.

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8 Close the Report dialog box.

Click Close.

Now that you are familiar with the process of designing steps for tests, you can proceed to Lesson 6, “Running Tests.” In Lesson 6, you will create a test set, and run manual and automated tests.

Tips

- Import existing WinRunner tests into your project database. Choose Plan > Import WinRunner Tests and specify the directory from which you want to import the tests. TestDirector places imported tests in the Unattached Tests folder in the test plan tree. You can then move these tests to any subject in the tree.
- View the tests in your project database in a grid by clicking the TestGrid icon. Each test appears as a record in the grid.
- Apply a filter to the test plan tree in order to display only tests that meet certain criteria. For example, you can choose to display only manual tests. To create a filter, choose View > Filter or click the Filter button.
- Attach files to a test, such as detailed specification documents or URLs. To attach a file to test, select the test in the test plan tree and click the Attach tab. Click File or Web to select the file.
- Change the order in which steps appear in the Design Steps grid. To reorder test steps, choose a test in the test plan tree and click the Design Steps tab. Click the Reorder button.

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Running Tests

After creating tests for each subject defined in the test plan tree, you are ready to run the tests. The method you use to run the tests depends on whether the tests are manual or automated.

This lesson includes:

- **What is a Test Set?**
- **Building a Test Set**
- **Running a Manual Test**
- **Running an Automated Test**
- **Analyzing Test Run**
- **Tips**

What is a Test Set?

After planning and creating a project with tests, you can start running the tests on your application. However, since a project database often contains hundreds or thousands of tests, deciding how to manage the test run process may seem overwhelming.

TestDirector helps you organize test runs by building *test sets*. A test set is a subset of the tests in your project, run together in order to achieve a specific goal. You build a test set by selecting tests from the test plan tree, and assigning this group of tests a descriptive name. You can then run the test set at any time, on any build of your application.

For example, consider building the following types of test sets:

Test Set	Purpose
Sanity	Tests the entire application at a basic level to check that it is stable.
Normal	Tests the system in a more in-depth manner than the sanity test. A Normal test set can contain both positive and negative checks. Positive checks test that the application responds to input as expected. Negative checks attempt to break the application.
Advanced	Checks the entire application, including its most advanced features.
Regression	Verifies that a change in one area of the application did not cause problems in other areas.
Function	Tests a specific module or function within the application.

In the following exercises you will create a Sanity test set that contains both automated and manual tests, and then run the tests from this set on the Flight Reservation application. Afterwards, you will create a graph which shows the progress of the test set execution.

Building a Test Set

In this exercise you will create a Sanity test set.

To create a test set:

1 **Open the TestDirector_Demo project.**

If TestDirector is not already open, choose TestDirector from the TestDirector 5.0 program group. The Login window opens. Select TestDirector_Demo from the Project list. Select admin from the User Name list. Click OK.

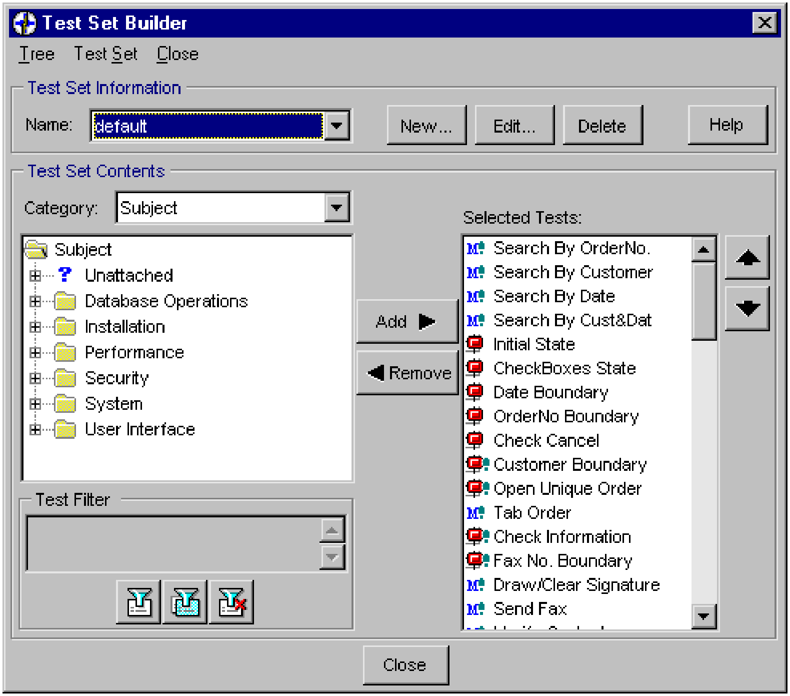
If TestDirector is already open, choose Project > Open. The Login window opens. Select TestDirector_Demo from the Project list. Select admin from the User Name list. Click OK.

2 **Display the Run Tests view.**

Click the Run Tests tab.

3 Open the Test Set Builder.

Choose Run > Test Set Builder, or click Test Set Builder. The Test Set Builder dialog box opens.



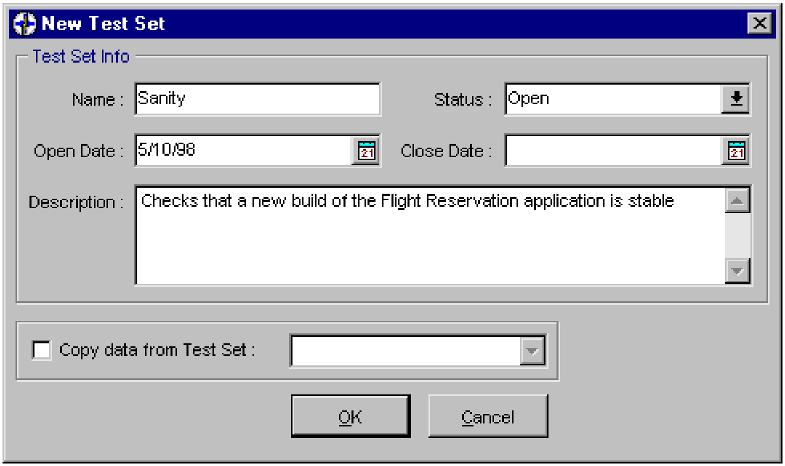
4 Create a new test set.

In the Test Set Builder, click New. The New Test Set dialog box opens.

In the Name box, type Sanity.

In the Status list, select Open. Click the check mark to close the list.

In the Description box, type Checks that a new build of the Flight Reservation application is stable.



5 Save the new test set.

Click OK.

6 Add Database Operation tests to the Sanity test set.

In the Test Set Builder, select the Database Operations folder in the test plan tree.

Click Add to add all the tests in the Database Operations folder to the test set.

7 Add User Interface tests to the test set.

Select the User Interface folder in the test plan tree.

Click Add to add all the tests in the User Interface folder to the test set.

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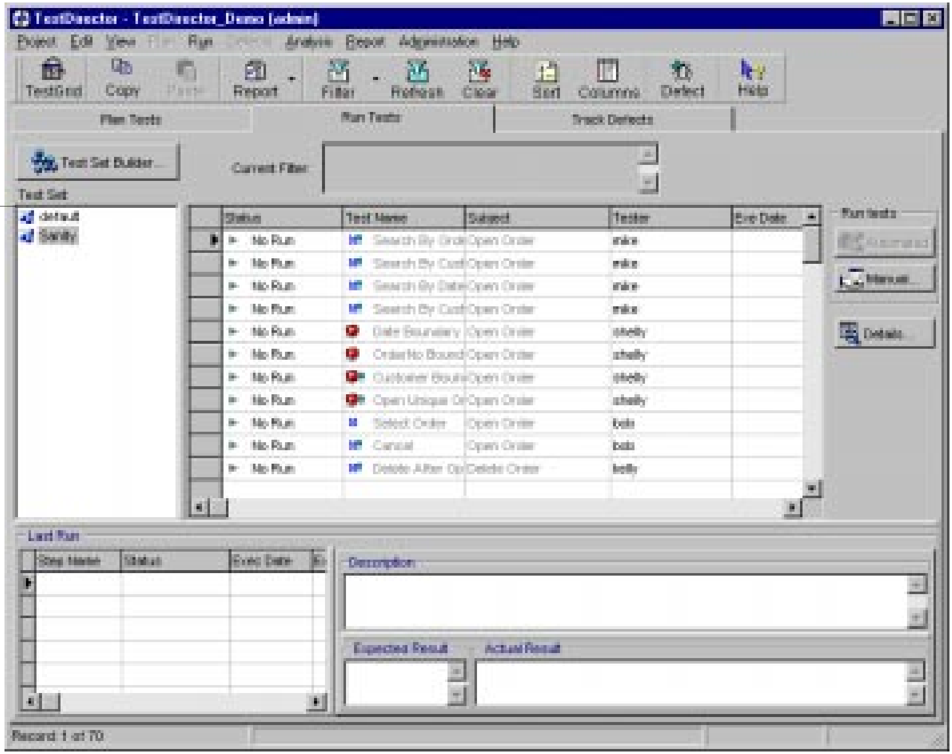
8 Save the Sanity test set and close the Test Set Builder.

Click Close. The test sets appear on the left side of the main grid in the Run Tests view.

Available test sets



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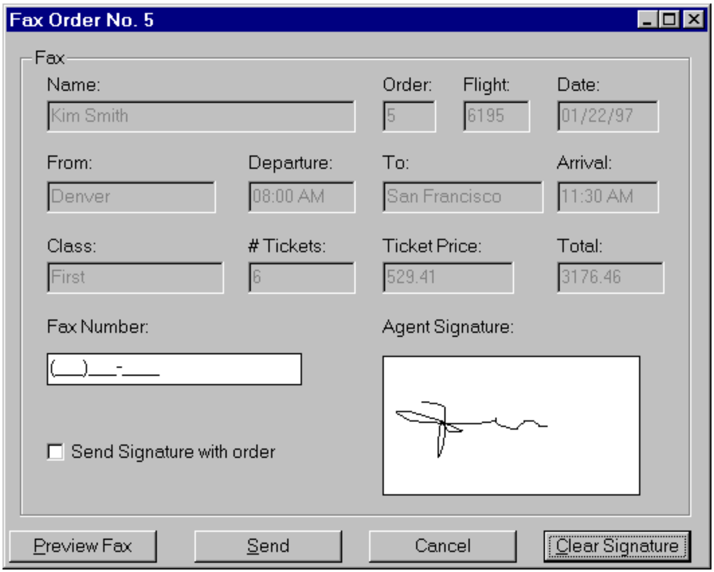
Running a Manual Test

After you create a test set, you can begin running tests on the application. You can perform many runs of the same test, each time saving the test run results under a different name.

In this exercise you will run the manual test Draw/Clear Signature. This test checks that you can draw and clear your signature in the Fax Order window of the Flight Reservation application (version 1B).



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Preparing the Manual Test Run

Before you start to run a manual test, you select the test in the test grid and fill in basic details about the test run such as the date, the tester name, and the name of the test run.

To prepare for a manual test run:

1 **Display the Run Tests view in TestDirector.**

If you are not in the Run Tests view, click the Run Tests tab.

2 **Display the Sanity test set**

In the Test Set list, select Sanity.

3 Locate the Draw/Clear Signature test.

Choose Edit > Find.

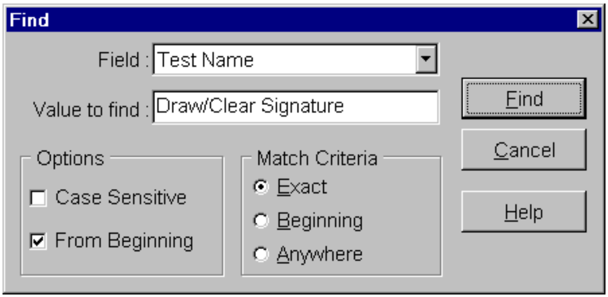
In the Field list, select Test Name.

In the Value To Find box, type Draw/Clear Signature.

Under Options, select From Beginning.

Under Match Criteria, click Exact.

Click Find.



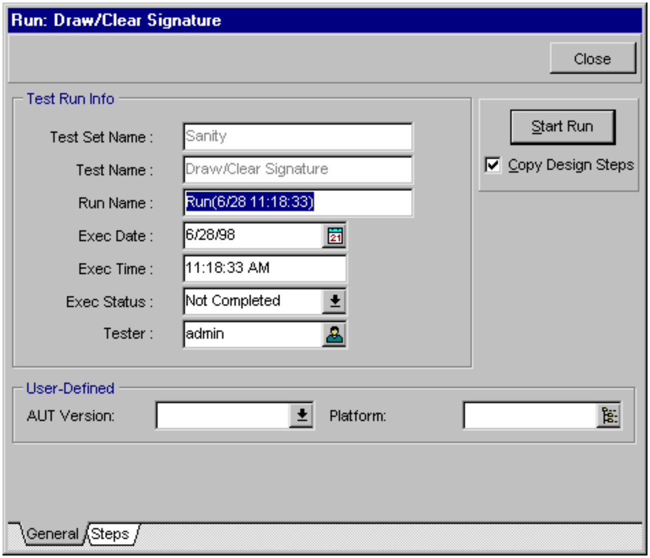
The Find command locates the Draw/Clear Signature test in the Run Tests grid.

4 Select the test.

In the Run Tests grid, click the gray box with the arrow to select the Draw/Clear Signature test.

5 Open the Run dialog box.

Click Manual. The main TestDirector window is minimized and the Run dialog box opens.



6 Enter details about the test run.

TestDirector automatically fills in the test run information.

In the AUT Version list, select Flight 1b. Click the check mark to close the list.

In the Platform list, select MS Windows. Click the check mark to close the list.

Make sure that the Copy Design Steps checkbox is selected.

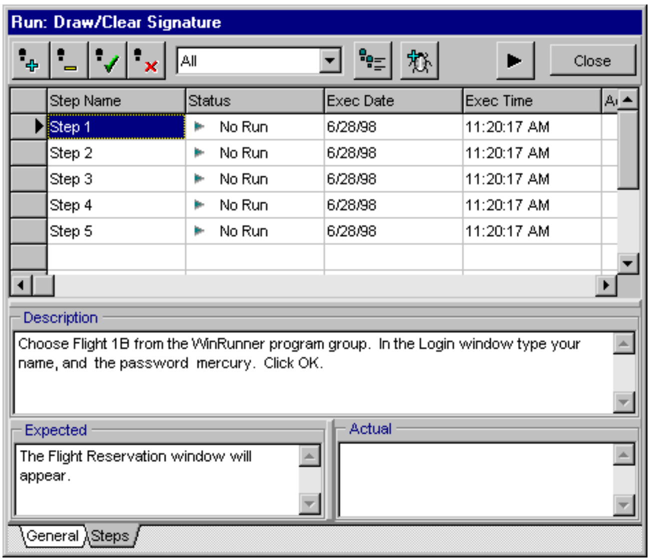
Starting the Manual Test Run

Once you have defined the basic information about the test, you are ready to start the test run.

To execute the manual test:

1 **Start the test run.**

Click Start Run. The Run dialog box opens displaying the test steps in a grid. The description and expected results of the selected step appear at the bottom of the dialog box.



Find
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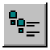
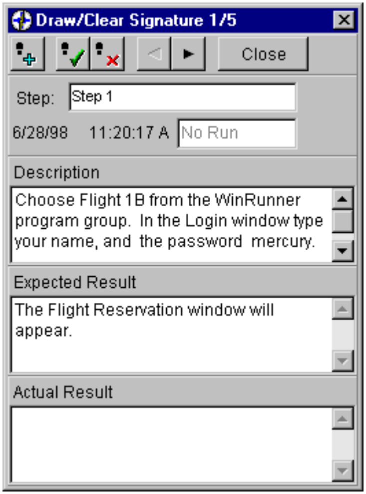


2 **Open the Mini Step window.**

Click the Mini Step button. The Run dialog box is minimized.



3 **Perform the first step.**

Perform the actions described in the Description box.

Read the expected result described in the Expected Results box.

In the Actual Results box, type The Flight Reservation window opens.

Click the Pass Step button.

4 **Perform the second step.**

Perform the actions described in the Description box.

Read the expected result described in the Expected Results box.

In the Actual Results box, type Order number 5 opens.

Click the Pass Step button.

5 **Perform the third step.**

Perform the actions described in the Description box.

Read the expected result described in the Expected Results box.

In the Actual Results box, type The Fax Order window opens.

Click the Pass Step button.

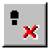
6 **Perform the fourth step.**

Perform the actions described in the Description box.

Read the expected result described in the Expected Results box.

In the Actual Results box, type Signature appears in the draw box.

Click the Pass Step button.



7 Perform the fifth step.

Perform the actions described in the Description box.

Read the expected result described in the Expected Results box.

In the Actual Results box, type Signature does not clear.

Click the Fail Step button.

8 Close the Mini Step window.

Click Close.

9 Close the Run window.

Click Close.

10 Close the Fax Order dialog box.

Click Cancel.

11 Close the Flight Reservation application.

Choose File > Exit.

Find

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Viewing the Test Run Results

You can view the results of the last test run in the Run Tests view. The updated status for the test run appears in the Status column of the Run Test grid.

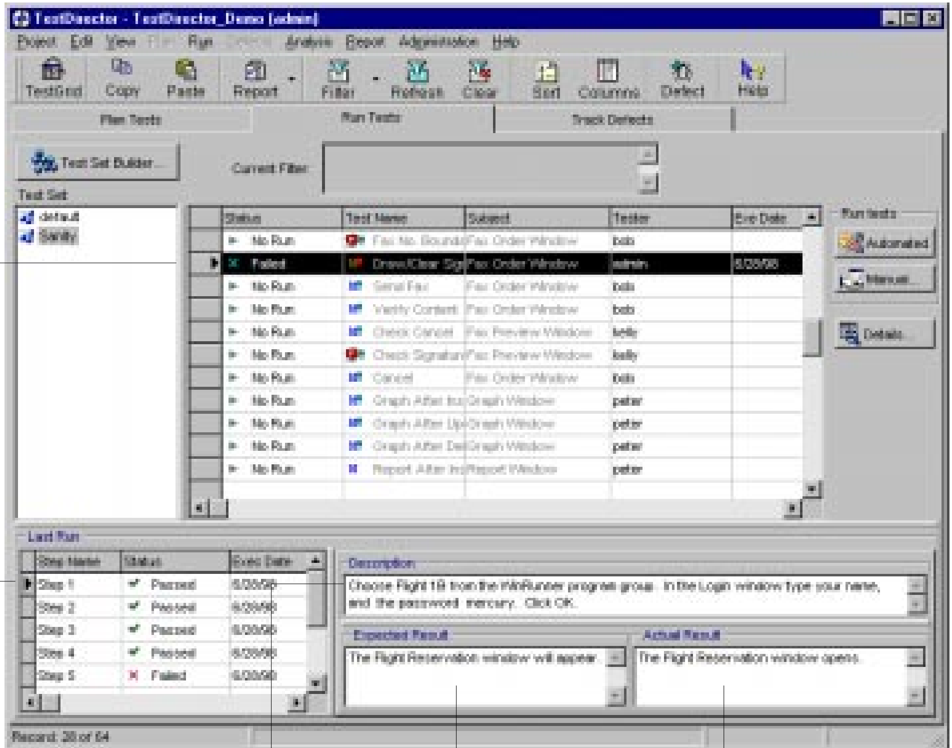
To view the results of the last test run:

1 **Make sure that the Last Run grid is displayed in the Run Tests view.**

If the Last Run grid is not displayed below the Run Tests grid, click View > Show Test Details.

Updated status of the test run

Results of each step



Description of a step

Expected result of a step

Actual result of a step

Find

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2 View the test steps.

Select the Draw/Clear Signature test in the Run Tests grid.

Click each step in the Last Run grid to view the step’s description, expected results, and actual results.

The Draw/Clear Signature test failed because of step 5. The Clear Signature button did not clear the signature box. In Lesson 7, “Tracking Defects”, you will report and track this defect.

Find

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Running an Automated Test

You can execute automated tests on your own computer, or on multiple remote hosts. A host is any computer connected to your network on which a testing tool has already been installed.

When you run a WinRunner test, TestDirector launches WinRunner. WinRunner loads and runs the test. After the test is completed, the test results are sent to the TestDirector project.

In order to run a WinRunner test from TestDirector, WinRunner must be installed on your computer (if you run the test locally). If you attempt to run an automated test and WinRunner does not open, consult your system administrator or Mercury Interactive's customer support.

Note: If you are using TestDirector standard edition, you can only run tests locally. The TestDirector client/server edition enables you to run tests both locally and on multiple remote hosts.

In this exercise, you will run the Insert New Order test locally. This test checks that ticket orders are inserted correctly into the Flight Reservation system.

Running a WinRunner Test

When the test run is finished, results appear in WinRunner's Test Results window, and in TestDirector's Run Test view.

To run an automated WinRunner test:

1 **Display the Run Tests view in TestDirector.**

If you are not in the Run Tests view, click the Run Tests tab.

2 **Display the Sanity test set**

In the Test Set list, select Sanity.

3 **Display only automated tests in the grid.**

To make it easier to locate the Insert New Order test, filter the tests in the grid so that only automated tests are displayed.

Click Filter. In the Filter dialog box, place the cursor inside the Filter Condition box of the "Plan: Type" box. Click the browse button.

In the Select Filter Definition dialog box, select WR-AUTOMATED and click OK.

Click OK to close the Filter dialog box. The automated WinRunner tests appear in the Run Tests grid.

4 Locate the Insert New Order test.

Using the scroll arrow, locate the Insert New Order test in the Test Name column of the Run Tests grid.

5 Select the test.

In the Run Tests grid, click the gray box with the arrow to select the Insert New Order test.

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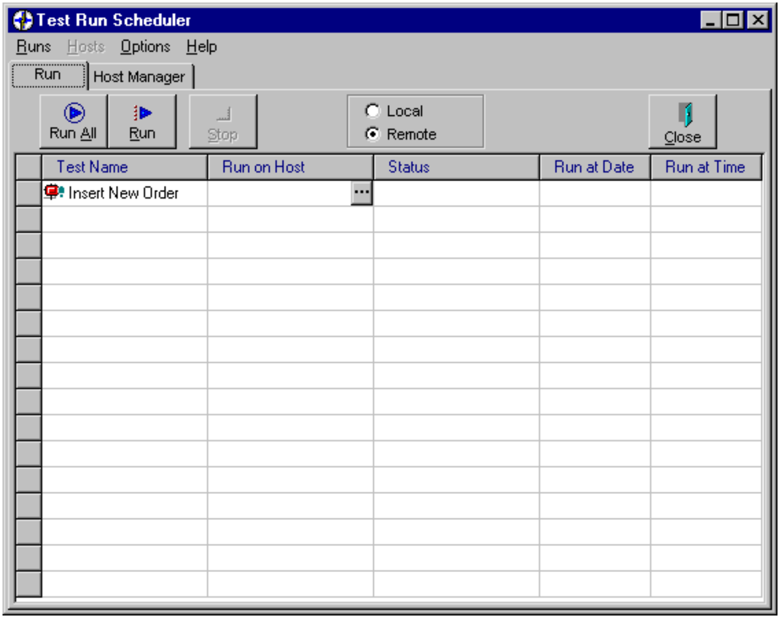
6 Open the Test Run Scheduler.

Click the Automated button. The Test Run Scheduler opens.



Note: In the TestDirector client/server edition, you can choose to run tests both locally and on remote machines. In the standard edition, the Local and Remote buttons are disabled since you can only run tests locally.

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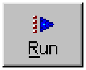
7 Set to run the test locally (TestDirector client/server edition only).

In the Test Run Scheduler, select Local to run the test on your computer. The name of your computer is displayed in the Run on Host column.



8 Run your test.

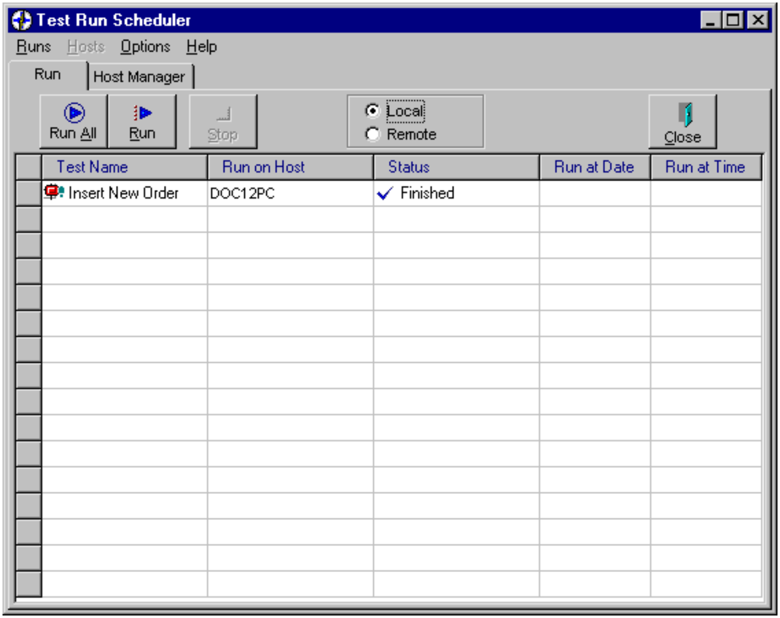
Click Run.

TestDirector opens the selected test in WinRunner and begins running the test on the Flight Reservation application. WinRunner compares the current response of your application to its expected response.

After test run is complete, WinRunner closes the test, but it remains open.

9 Verify the status of the test run in the Test Run Scheduler.

Click the Test Run Scheduler task button.



The Status column indicates that the test run is finished. Click Close.

After the test run is complete, you can view the test results in the WinRunner Test Results window, and in TestDirector in the Run Test view.

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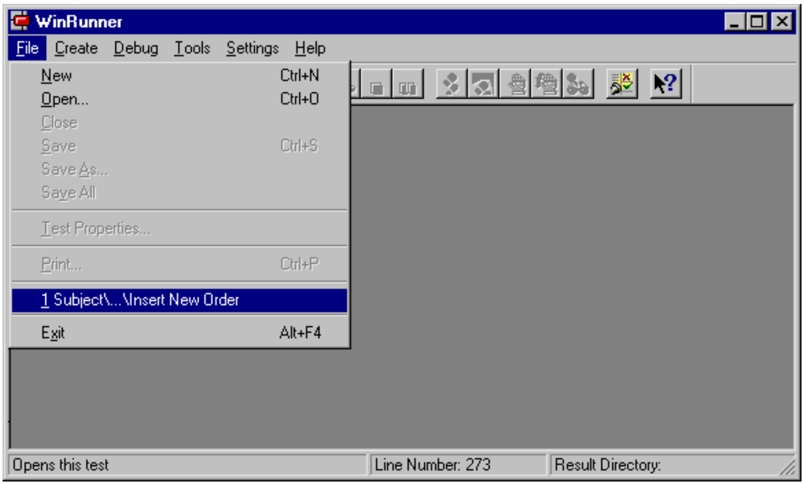
Viewing the Test Run Results in WinRunner

Once a test run is completed, you can view detailed test results in WinRunner's Test Results window. WinRunner color-codes results (green=passed, red=failed) so that you can quickly draw conclusions about the success or failure of the test.

To view the test results in WinRunner:

1 **Open the test.**

Choose File > Insert New Order.



The Insert New Order test opens.

2 Open the WinRunner Test Results window.

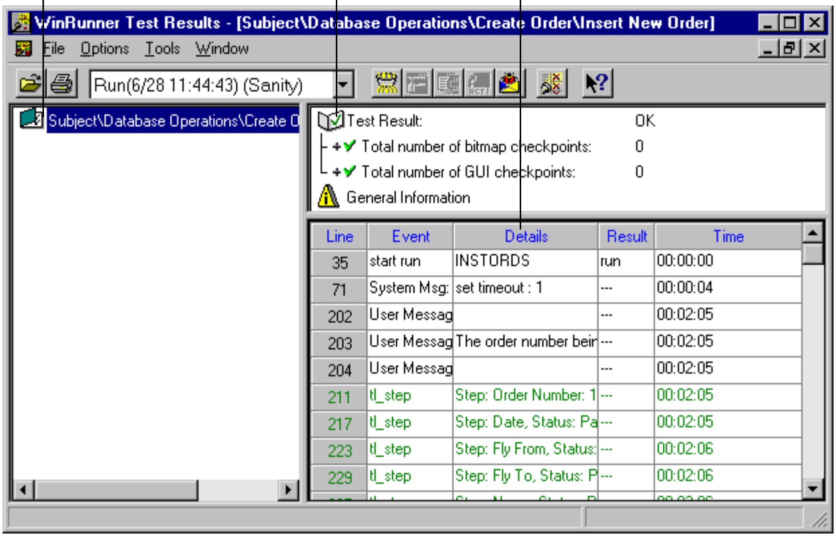
Choose Tools > Test Results, or click the Test Results button. The WinRunner Test Results window opens.

1 2 3

1 Test Tree: Shows all tests executed during the test run.

2 Test Summary: Shows whether a test run passed or failed. It also lists general information about the test run.

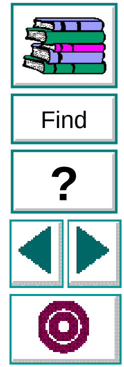
3 Test Log: Lists the major events that occurred during the test run. It also lists the test script line at which each event occurred.



3 View the test results.

The Test Summary section indicates that the test result is OK.

For more information on viewing test results in WinRunner, refer to the *WinRunner User's Guide*.



4 Close the report.

Choose File > Exit.

5 Close WinRunner.

Choose File > Exit in the WinRunner menu.

Click No if a WinRunner message box appears asking you to save changes.

6 Close the Flight Reservation application.

Choose File > Exit.

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Viewing the Test Run Results in TestDirector

TestDirector displays test run results in the Run Tests view and in the Run Test-Details dialog box.

To view the test results in TestDirector:

1 View the results in the Run Test grid.

Click Refresh to view the latest test results in the Run Tests grid. The status for the test run is updated from “No Run” to “Passed”.

2 View the results in the Last Run grid.

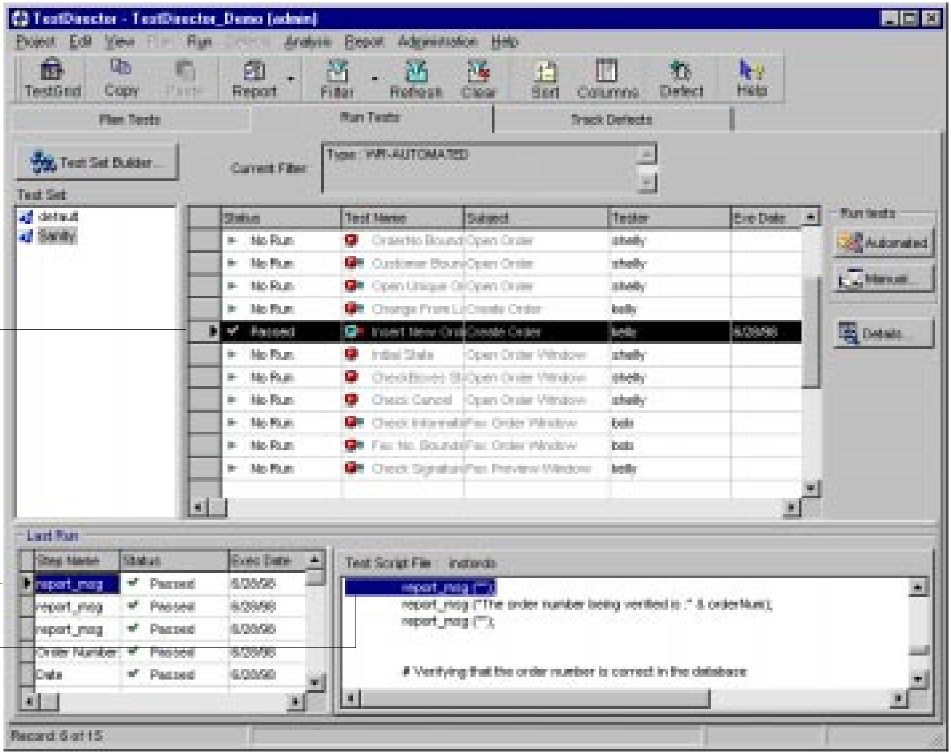
If the Last Run grid is not displayed, click View > Show Test Details. The Last Run grid appears below the Run Tests grid.

View the test script line for a step by clicking the step name. The test script line for the selected step is highlighted in the Test Script File section.

Updated status of the test run

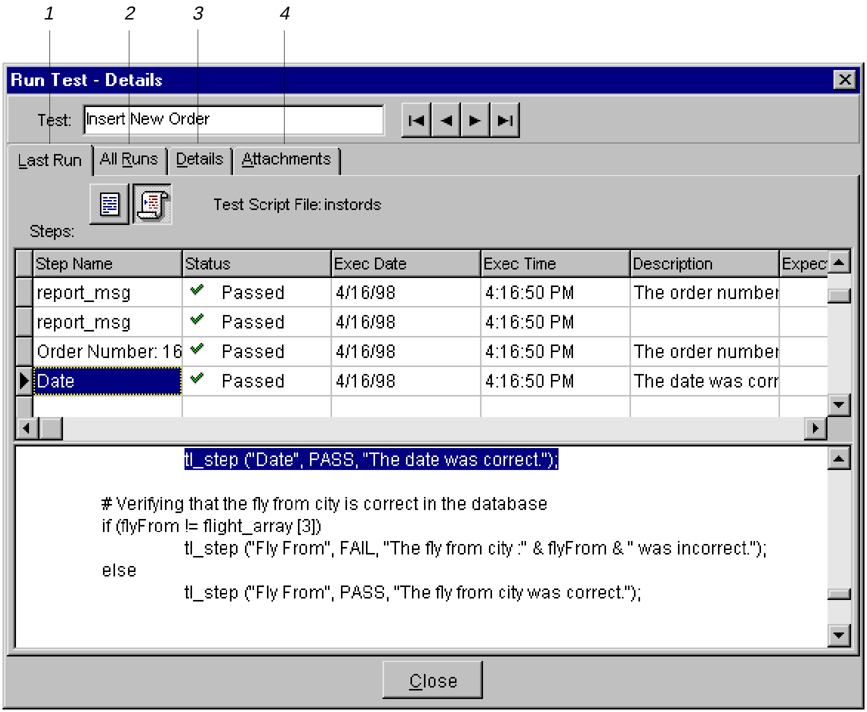
Step name

Test script line



3 View the results in the Run Test-Details dialog box.

Click Details. The Run Test Details dialog box opens.



1 *Last Run tab: Shows the results for each step of the most recent test run.*

2 *All Runs tab: Shows step results for all the runs of a selected test.*

3 *Details tab: Shows a description of the selected test and provides test run statistics.*

4 *Attachments tab: Shows any additional files attached to the test.*

4 Close the Run Test-Details dialog box.

Click Close.

Analyzing Test Run

After running a test, you can evaluate the progress of the test run by creating a graph. A graph can help you draw conclusions quickly and see the relationships between the different types of test data.

In this exercise you will create a summary graph showing the execution status for all the manual tests.

To create a graph:

1 **Display the Run Tests view.**

Click the Run Tests tab.

2 **Create a filter to view manual tests only.**

In the Run Test view, click Filter.

In the Filter dialog box, place the cursor inside the Filter Condition box of the “Plan: Type” box. Click the browse button.

In the Select Filter Definition dialog box, select MANUAL and click OK.

Click OK to close the Filter dialog box. The manual tests appear in the Run Tests grid.

3 **Select a graph.**

Choose Analysis > Execution Summary Graphs.

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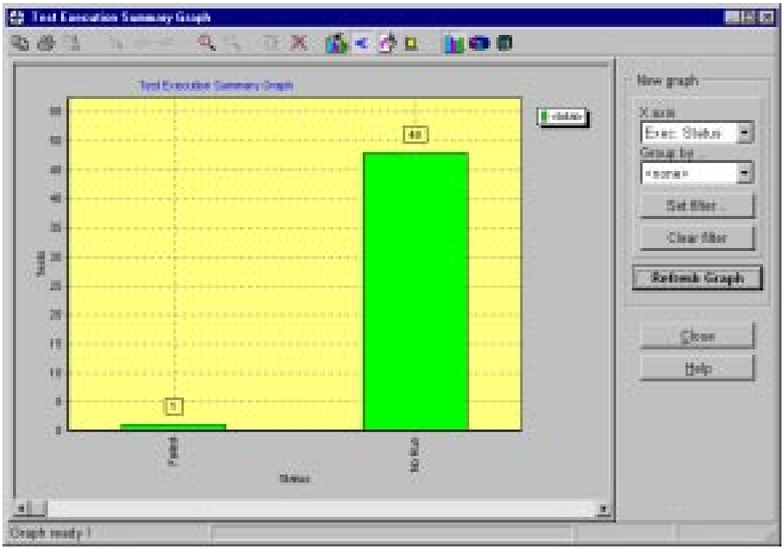
4 Customize the graph.

In the graph window, select “Exec: Status” from the X axis list. In the Group by list, select “<none>”.

5 Display the new graph.

Click Refresh Graph.



In the example above, the graph indicates that out of 49 manual tests, 1 test failed (Draw/Clear Signature test).

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6 Close the graph.

Click Close.

Now that you are familiar with the process of creating a test set and running manual and automated tests, you can proceed to Lesson 7, “Tracking Defects.” In Lesson 7, you will report and track the defect detected in the Draw/Clear Signature test.

Tips

- If a manual test run is interrupted during execution, you can continue the run later on. Choose Run > Continue Manual Test. The Run dialog box opens, and you continue executing the remaining steps in the test.
- You can run multiple automated tests in WinRunner using TestDirector. You select the tests to run, and TestDirector creates a batch test for them. This test opens and executes each test in the batch, and saves the results in WinRunner. In the Run Tests grid, select the automated tests that you want to run. Choose Run > Create WinRunner Batch Test, and assign a name to the group. You execute the batch test using the same method as a regular automated test.
- To run all of the automated tests in a test set, click Run > Run All Automated Tests.
- You can use the Remote Run view in the Test Run Scheduler, to run tests on multiple remote hosts.
- You can set the date and time you want your test to run in the Test Run Scheduler. Click the Run at Date column header and type the date you want the test to run. To set the time, click the Run at Time column header.
- You can use the Host Manager view in the Test Run Scheduler, to manage hosts. For example, you use the view to create a list of hosts, verify that hosts are capable of running tests, and select which hosts will not run tests. You also use the Host Manager view to create groups of hosts. For example, you could create a group that contains all the Windows 95 hosts you want to use for a particular project.

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- You can compare the results of one test run to the results of previous runs. Using the Test Run Details dialog box, you can view run details and step results for all the runs of a selected test.
- You can remove the accumulated test run results from a test set. You can then rerun the tests, viewing only the new test run data. In the Run Tests view, select the test set whose run data you want to delete. Choose Run > Reset Test Set. A message appears verifying your decision.

Tracking Defects

Once you detect a defect in the software, you need to report and track it until it is repaired.

This lesson includes:

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How Do You Keep Track of Defects?

Locating and repairing software defects is an essential phase in software development. Defects can be detected and reported by software developers, testers, and end users in all stages of the testing process. Using TestDirector, you can report flaws in your application, and track data derived from defect reports.

When a defect is detected in the software:

- ✔ Send a defect report to the TestDirector database.
- ✔ Review the defect and assign it to a member of the development team.
- ✔ Repair the open defect.
- ✔ Test a new build of the application after the defect is corrected. If the defect does not reoccur, change the status of the defect.
- ✔ Generate reports and graphs to help you analyze the progress of the defects in your TestDirector project.

In Lesson 5, “Running Tests” the Draw/Clear Signature test failed. The Clear Signature button did not clear the signature box. In the following exercises, you will report this new defect to the project database. You will then track the repair process of the defect.

Reporting a New Defect

You can report a new defect at any stage of the testing process by adding a defect record to the project database. Each defect is tracked through four stages: *New*, *Open*, *Fixed*, and *Closed*. When you initially report a defect to the project database, you assign it the status *New*.

In this exercise you will use the New Defect dialog box to report that the Clear Signature button did not clear the signature box in the Draw/Clear Signature test.

To report the new defect:

1 **Open the TestDirector_Demo project.**

If the TestDirector_Demo project is not already open, choose Project > Open from TestDirector. The Login window opens. Select TestDirector_Demo from the Project list. Select admin from the User Name list. Click OK.

2 **Display the Track Defects view.**

Click the Track Defects tab.

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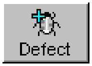
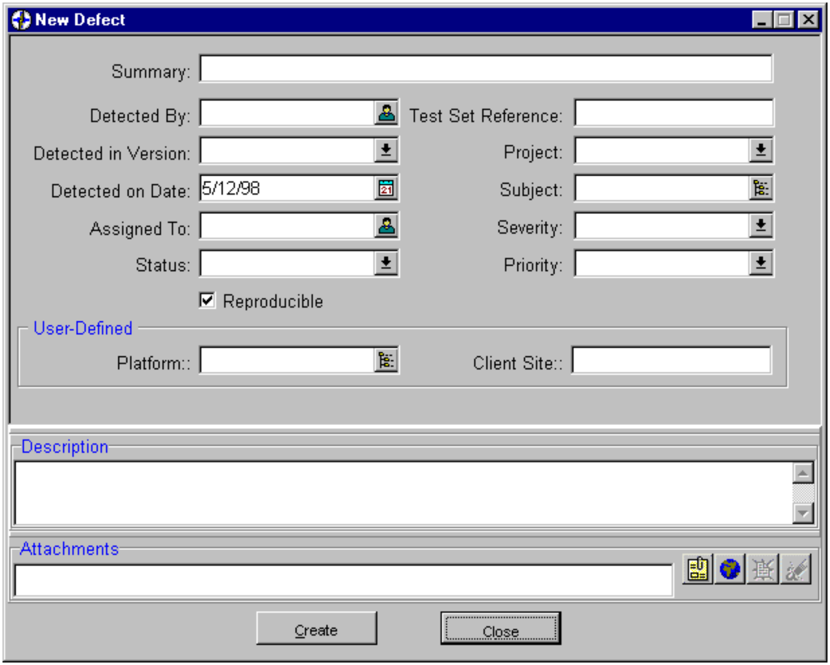


3 Open the New Defect dialog box.

Click Defect.



4 Type a short description of the defect.

In the Summary box, type a short description of the defect. For example, type Clear Signature button doesn't work.

5 **Specify basic defect information.**

In the Detected By list, specify the name of the person who detected the defect. Select Alec. Click the check mark to close the list.

In the Detected in Version list, specify the software version in which the defect was detected. Select Flight 1b. Click the check mark to close the list.

Skip the Assigned To field. You will assign the person responsible for tracking the defect when you change the status of the new defect to Open.

In the Status list, select New. Click the check mark to close the list.

In the Test Set Reference box, type Sanity.

In the Project list, select Project 1. Click the check mark to close the list.

In the Subject list, double-click the folder User Interface, and select Fax Order Window. Click the check mark to close the list.

6 **Determine the degree to which the defect affects the application.**

In the Severity list, select 4-Very High. Click the check mark to close the list.

7 **Determine when the defect will be repaired in relation to other defects in the project database.**

In the Priority list, select 3-High. Click the check mark to close the list.

8 Indicate whether the defect can be reproduced under the same conditions by which it was detected.

Select the Reproducible check box.

9 Skip the User-Defined area.

The Platform and Client Site are user-defined fields. For the purpose of this exercise, skip these fields.

10 Type a detailed description of the defect.

In the Description box, type a description of the defect. For example, type The defect was detected in step 5 of the Draw/Clear Signature test. The Clear Signature button did not clear the signature box.

11 Skip the Attachments field.

You can attach a file to a defect record. This is particularly useful for referencing relevant files such as detailed defect descriptions and captured screen images. Refer to the *TestDirector User's Guide* for further information on attachments.

For the purpose of this exercise, skip this field.

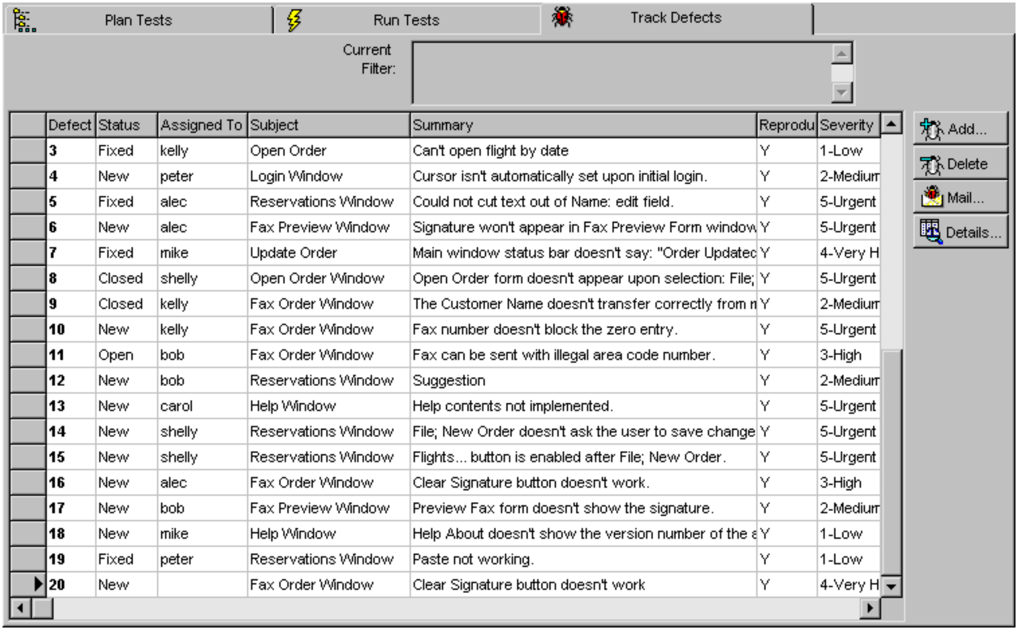
12 Create a defect record.

Click Create to save the defect record in the project database. A message appears indicating that the defect number 20 was added successfully. Click OK.

Click Close to exit the New Defect dialog box.

13 View the new defect record in the Track Defects grid.

Locate defect number 20 in the Track Defects grid. Note that the defect status is New.



Defect	Status	Assigned To	Subject	Summary	Reprodu	Severity
3	Fixed	kelly	Open Order	Can't open flight by date	Y	1-Low
4	New	peter	Login Window	Cursor isn't automatically set upon initial login.	Y	2-Medium
5	Fixed	alec	Reservations Window	Could not cut text out of Name: edit field.	Y	5-Urgent
6	New	alec	Fax Preview Window	Signature won't appear in Fax Preview Form window	Y	5-Urgent
7	Fixed	mike	Update Order	Main window status bar doesn't say: "Order Updatec	Y	4-Very H
8	Closed	shelly	Open Order Window	Open Order form doesn't appear upon selection: File;	Y	5-Urgent
9	Closed	kelly	Fax Order Window	The Customer Name doesn't transfer correctly from n	Y	2-Medium
10	New	kelly	Fax Order Window	Fax number doesn't block the zero entry.	Y	5-Urgent
11	Open	bob	Fax Order Window	Fax can be sent with illegal area code number.	Y	3-High
12	New	bob	Reservations Window	Suggestion	Y	2-Medium
13	New	carol	Help Window	Help contents not implemented.	Y	5-Urgent
14	New	shelly	Reservations Window	File; New Order doesn't ask the user to save change	Y	5-Urgent
15	New	shelly	Reservations Window	Flights... button is enabled after File; New Order.	Y	5-Urgent
16	New	alec	Fax Order Window	Clear Signature button doesn't work.	Y	3-High
17	New	bob	Fax Preview Window	Preview Fax form doesn't show the signature.	Y	2-Medium
18	New	mike	Help Window	Help About doesn't show the version number of the a	Y	1-Low
19	Fixed	peter	Reservations Window	Paste not working.	Y	1-Low
20	New		Fax Order Window	Clear Signature button doesn't work	Y	4-Very H

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Updating a Defect Record

Tracking the repair of defects in a project requires that you periodically update defect records until they are resolved. You monitor the defect repair process by viewing defect records in the Track Defects grid and Defect Details dialog box.

To view the defect details:

1. **Display the Track Defects view in TestDirector.**

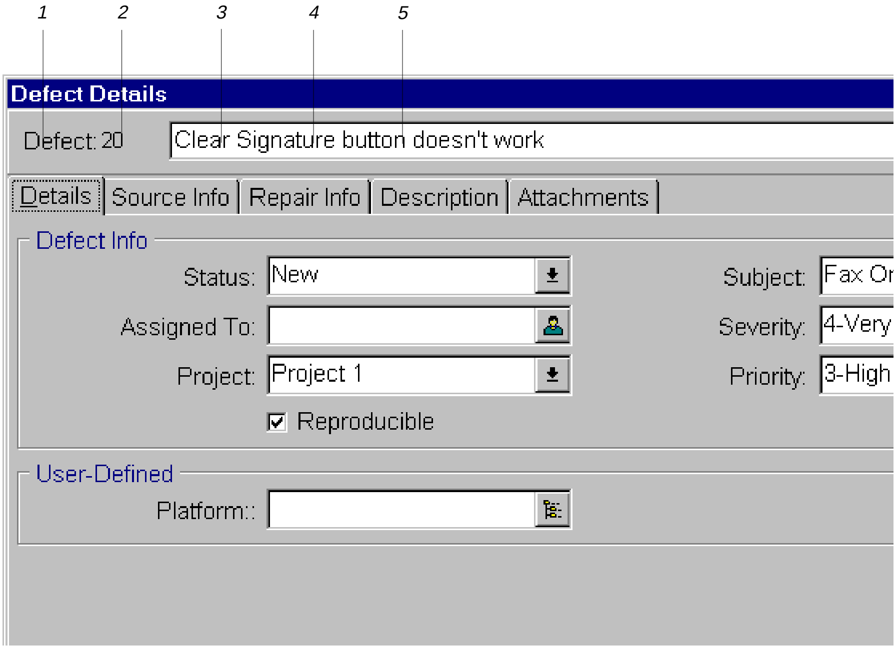
   If you are not in the Track Defects view, click the Track Defects tab.

2. **Select the defect record.**

   In the Defect grid, select defect number 20 “Clear Signature button doesn’t work”.

3 View the details of the defect.

Click Details. The Defect Details dialog box appears, displaying information on the currently selected defect.



1 *Details view: Displays basic defect information.*

2 *Source Info view: Displays information on defect detection.*

3 *Repair Info view: Displays information on the planned and actual closing of a defect.*

4 *Description view: Displays the defect description and R&D comments.*

5 *Attachments view: Displays the file(s) associated with a defect.*

Tracking the Defect

When you initially reported defect number 20, you assigned it the status *New*. A quality assurance or project manager reviews the new defect, changes its status to *Open*, and assigns it to a member of the development team.

A developer identifies the cause of the defect, modifies the application script, and recompiles the application. When the defect is repaired, it is assigned the status *Fixed*.

The test is executed on the new build of the application. If a defect does not reoccur, the quality assurance or the project manager assign it the status *Closed*.

In this exercise you will update defect number 20, using the Defect Details dialog box. You will change the defect status from *New* to *Open*, and assign the defect to Shelly.

To update the defect status to Open:

1 **Display the Details view.**

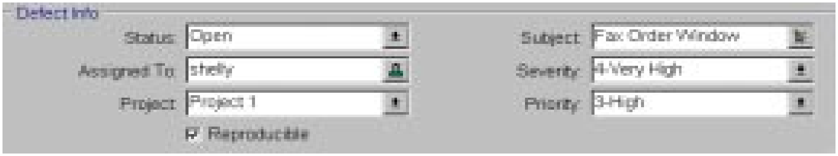
Click the Details tab.

2 **Change the status of the defect.**

In the Status list, select Open. Click the check mark to close the list.

3 Assign the defect to a member of the development team.

In the Assigned To list, select Shelly. Click the check mark to close the list.



4 Close the Defect Details dialog box.

Click Close to save the updated information in the project database.

5 View the updated information in the Track Defects grid.

Locate the defect number 20 in the grid. Scroll the grid to verify the updated information for the defect. Note that the defect status is Open.

Analyzing Defect Tracking

In this exercise, you will create a summary graph that shows any new or open defects with the repair priority of 3-High or higher.

To create a graph:

1 **Display the Track Defects view.**

Click the Track Defects tab.

2 **Select a graph.**

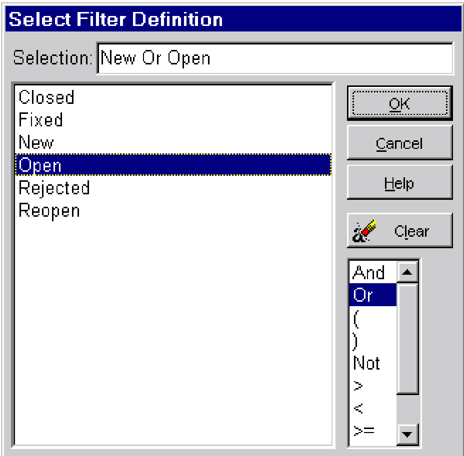
Choose Analysis > Defect Summary Graphs.

3 Create a filter to view new or open defects.

In the Defect Summary Graph dialog box, click Set Filter.

In the Filter dialog box, place the cursor inside the Filter Condition box of the Status box. Click the browse button.

In the Select Filter Definition dialog box, select New from the list. Select Or to create a logical expression. Select Open from the list.



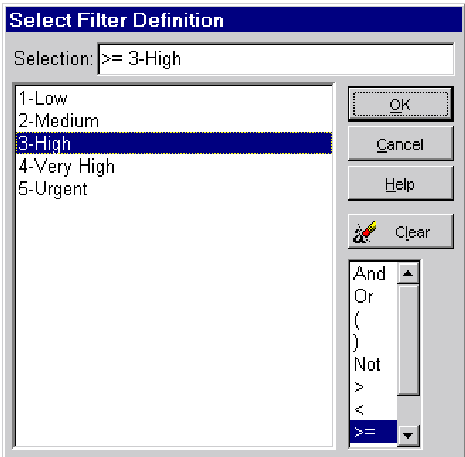
4 Close the Select Filter Definition dialog box.

Click OK.

5 Create a filter to view defects with high to urgent priority.

In the Filter dialog box, place the cursor inside the Filter Condition box of the Priority box. Click the browse button.

In the Select Filter Definition dialog box, select the logical expression >=. Select 3-High from the list.



6 Close the Select Filter Definition dialog box.

Click OK.

7 Close the Filter dialog box.

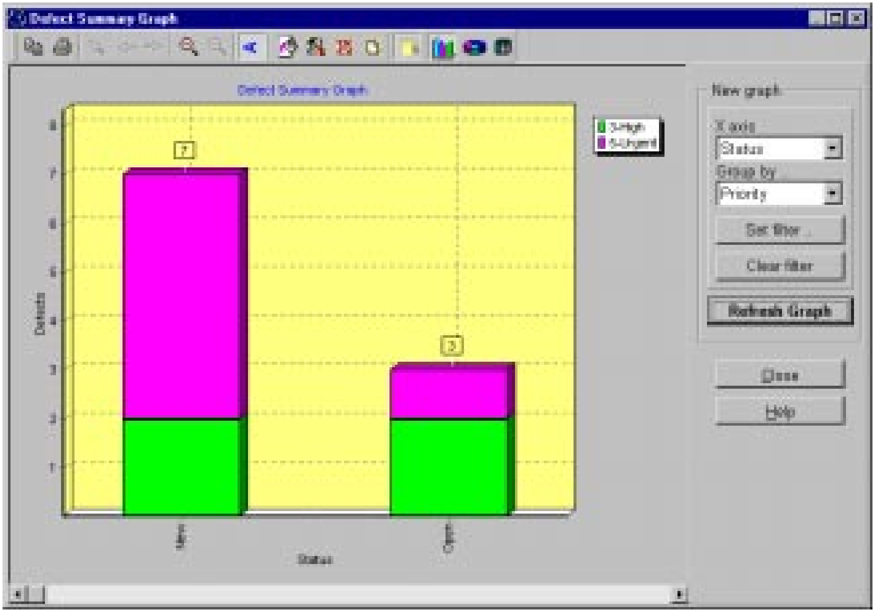
Click OK.

8 Customize the graph.

In the X axis list, select Status. In the Group by list, select Priority.

9 Display the new graph.

Click Refresh Graph.



Find
?

In the example above, the New column indicates that out of 7 new defects, 5 defects are assigned the priority 5-Urgent and 2 defects are assigned the priority 3-High.

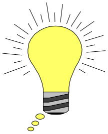
The Open column indicates that out of the 3 open defects, 1 defect is assigned the priority 5-Urgent, and 2 are assigned 3-High.

Based on the graph, you can conclude that 10 defects have the repair priority of high to urgent and should be fixed immediately.

10 **Close the graph.**

Click Close.

Find
?



Tips

- You can send a defect report using TestDirector or using the Remote Defect Reporter. The Remote Defect Reporter enables development and testing personnel to report defects without logging on to TestDirector. You can invoke the Remote Defect Reporter as a stand-alone application or by the Report dialog box in WinRunner. For more information, refer to the *Remote Defect Reporter User's Guide*.
- Connecting TestDirector to your e-mail system lets you inform development and quality assurance personnel routinely about defect repair activity. You determine the conditions for sending defect messages to each recipient by defining a mailing configuration. For more information, refer to the "Defect Tracking" section in the *TestDirector User's Guide*.

Summary

TestDirector guides you through the test planning, test running, and defect tracking phases of the testing process. This chapter provides you with a review of how to manage the software testing process for your application.

Test Management Process

The test management process involves three main phases:

- *Planning Tests:* Divide your application into test subjects and build a database of tests (project).
- *Running Tests:* Create test sets and perform test runs.
- Tracking Defects: Report defects detected in your application and track how repairs are progressing.

Throughout each phase, you can analyze data by generating detailed reports and graphs.

Planning Tests

Many quality experts believe that planning is the most important stage of the testing process. With a good test plan, you can assess the quality of your application at any time.

The Planning Tests process consists of six main stages:

Tasks	Description
Define Testing Goals	Examine your application, system environment, and testing resources in order to determine your testing goals.
Define Test Subjects	Divide your application into modules or functions to be tested. Build a *test plan tree* to hierarchically divide your application into testing units, or *subjects*.
Define Tests	Determine the types of tests you need for each module. Add a basic definition of each test to the test plan tree.
Design Test Steps	Develop manual tests by adding steps to the tests in your test plan tree. *Test steps* describe the test operations, the points to check, and the expected outcome of each test. Decide which tests to automate.

Tasks	Description
Automate Tests	For tests that you decide to automate, create TSL scripts with WinRunner, Mercury Interactive's automated GUI testing tool for Windows environments.
Analyze Test Plan	Generate reports and graphs to assist in analyzing test planning data. Review your tests to determine their suitability to your testing goals.

Running Tests

Test Execution is the heart of the testing process. Each time your application changes, you will want to execute the relevant parts of your test plan in order to locate defects and assess quality.

The Running Tests process consists of three main stages:

Tasks	Description
Create Test Set	Define groups of tests to meet the various testing goals in your project. These might include, for example, testing a new software version or a specific function in an application. Determine which tests to include in each test set.
Run Test Set	Schedule test execution and assign tasks to software testers. Execute the manual and/or automated tests in a set.
Analyze Test Results	View the results of your test runs in order to determine whether a defect has been detected in your application. Generate reports and graphs to help you analyze these results.

Tracking Defects

Locating and repairing software defects is an essential part of software development. Defects can be detected and reported by engineers, testers, and end-users in all phases of the testing process. Information about defects must be detailed and organized in order to schedule defect fixes and determine software release dates.

Defect Tracking involves three main stages:

Tasks	Description
Report Defects	Report new defects detected in your application. Quality assurance testers, developers, project managers, and end users can report defects during any phase of the testing process.
Track Defects	Review new defects and determine which ones should be fixed. Correct the defects that you decided to fix. Test a new build of your application. Continue this process until defects are repaired.
Analyze Defect Data	Generate reports and graphs to assist you in analyzing the progress of defect repairs, and to help you determine when to release the software.

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TestDirector Tutorial

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